

Practical considerations

How do you find the most parsimonious tree?

Table 3.1: The number of rooted, bifurcating, labeled trees for n species, for various values of n . The numbers for more than 20 species are approximate.

One way:

1. Try every possible tree.
2. For each tree, count min # of steps
3. Choose the tree with the fewest # steps

Too many trees.

Species	Number of trees
1	1
2	1
3	3
4	15
5	105
6	945
7	10,395
8	135,135
9	2,027,025
10	34,459,425
11	654,729,075
12	13,749,310,575
13	316,234,143,225
14	7,905,853,580,625
15	213,458,046,676,875
16	6,190,283,353,629,375
17	191,898,783,962,510,625
18	6,332,659,870,762,850,625
19	221,643,095,476,699,771,875
20	8,200,794,532,637,891,559,375
30	4.9518×10^{38}
40	1.00985×10^{57}
50	2.75292×10^{76}

Finding the best tree: Exhaustive searches

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Heuristic searches

Don't try every tree.

"Hill-climbing algorithm"

Wander around under certain rules.
Hope that what you eventually stop
at is the best.

Not guaranteed to find the best tree.

Can be improved by lots of random
starting points.

(Not necessary for parsimony, exact
algorithmic solution possible with a
"branch and bound" search)

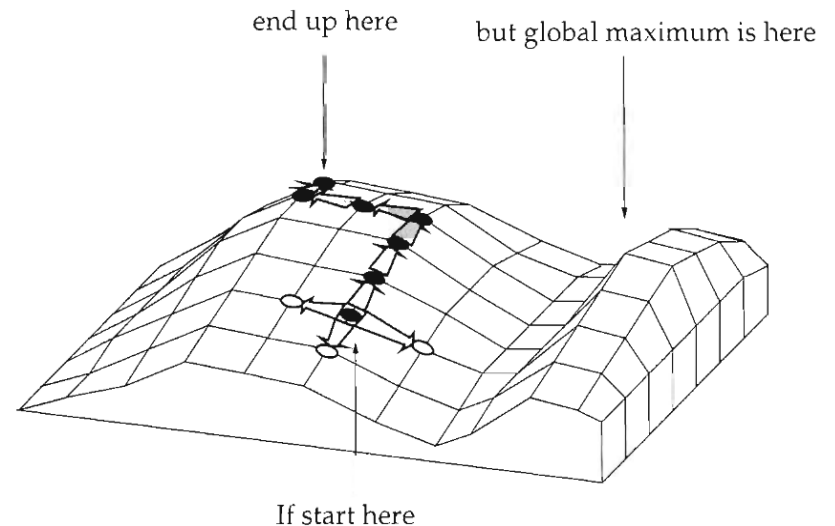


Figure 4.1: A surface rising above a two-dimensional plain (or plane). The process of climbing uphill on the surface is illustrated, as well as the failure to find a higher peak by this "greedy" method.

Tree space

Each point on the grid is a possible tree.

Trees next to each other are "1 step away" via algorithmic manipulation

Start at a random tree.

Try all trees that are nearby, choose the one that's best

Keep doing that until you can't make the tree any better.

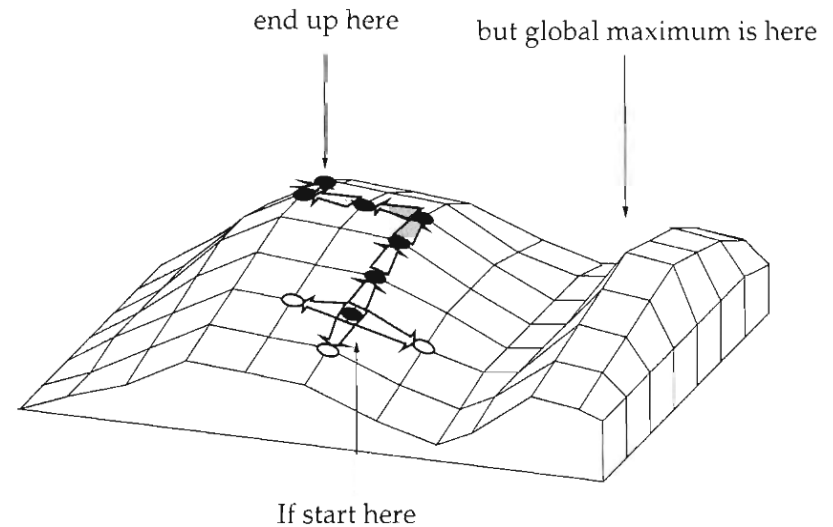


Figure 4.1: A surface rising above a two-dimensional plain (or plane). The process of climbing uphill on the surface is illustrated, as well as the failure to find a higher peak by this "greedy" method.

Visualizing tree space

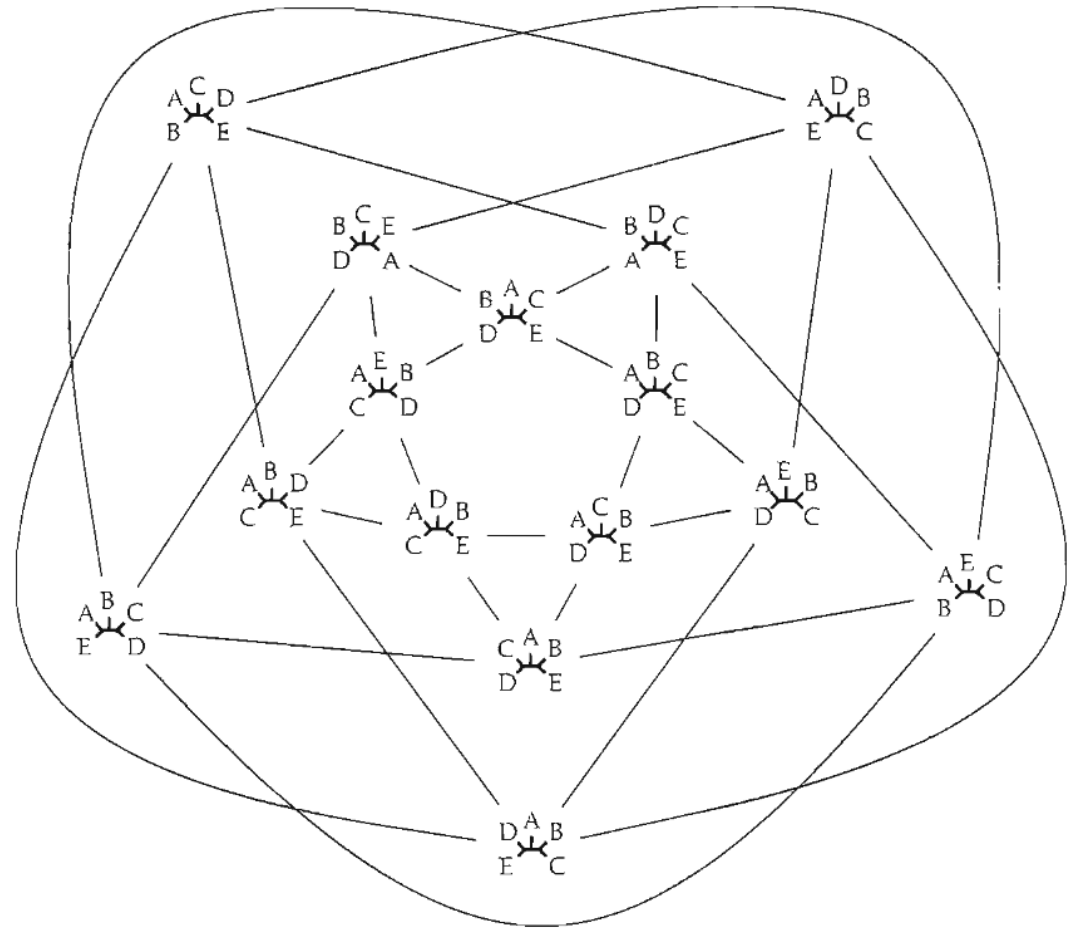


Figure 4.3: The space of all 15 possible unrooted trees with 5 tips. Neighbors are connected by lines when a nearest-neighbor interchange can convert one into the other. The labels A–E correspond to the species names Alpha through Epsilon in that data set. This symmetric arrangement of nodes was discovered by Ben Rudd Schoenberg (personal communication), and we thus denote this graph the Schoenberg graph.

Finding the shortest tree in treespace

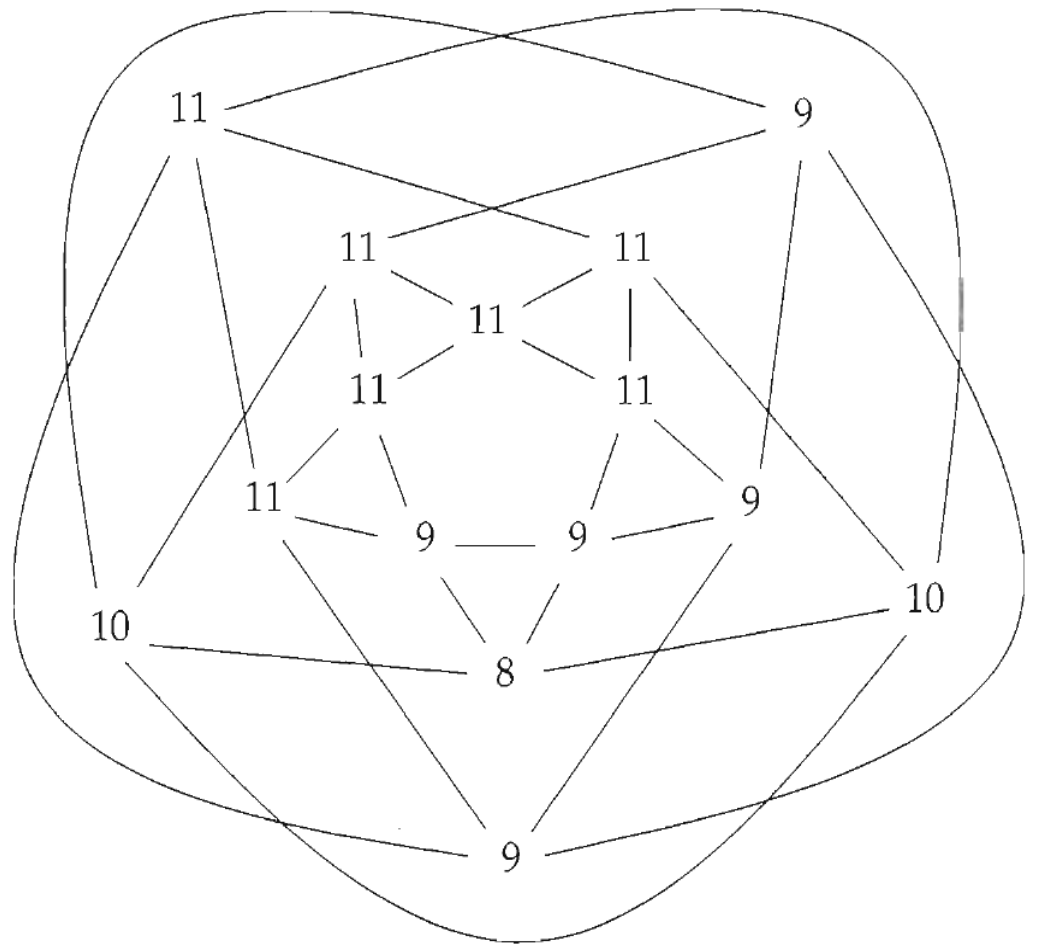
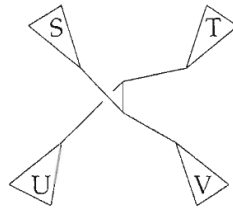
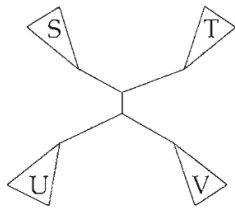


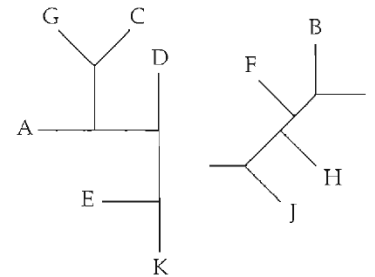
Figure 4.4: The space of all 15 possible trees, as in Figure 4.3, where the number of changes of state on the data set of Table 1.1 is shown. Nearest-neighbor interchanges search for the most parsimonious tree by moving in this graph.

Tree space connectivity is defined by "moves"

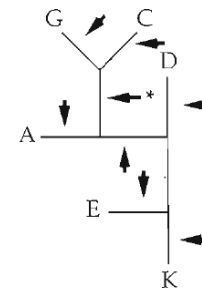


Nearest Neighbor Interchange (NNI)

Break a branch, remove a subtree

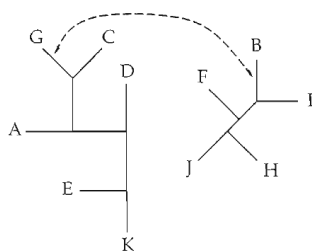


Add it in, attaching it to one (*) of the other branches



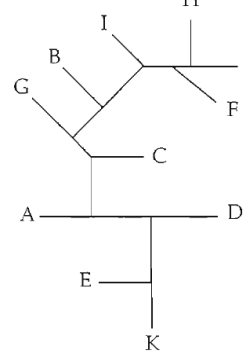
Subtree pruning and regrafting (SPR)

Connect a branch of one to a branch of the other

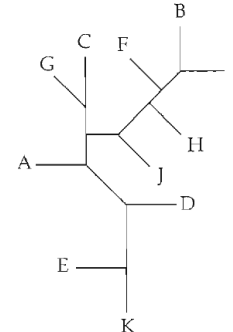


Tree bisection and reconnection (TBR)

Here is the result:



Here is the result:



Distance between tree spaces
changes depending on which
moves you use

NNI fastest, least computationally intensive, but
the largest distances between trees

TBR slowest, most computationally intensive, but
shortest distances between trees

SPR intermediate

Navigating tree space

Optimization/gradient descent (here ascent) is a general problem in computational analyses

It is always a balance of speed (NNI faster than SPR faster than TBR) and how efficiently you move across tree space (TBR connects more trees than SPR which is more than NNI)

Unless you sample all trees, a heuristic search may not be sure you ever find the global peak.

Trying lots of different starting points can increase your confidence.

Parsimony has an exact algorithm

Branch and bound

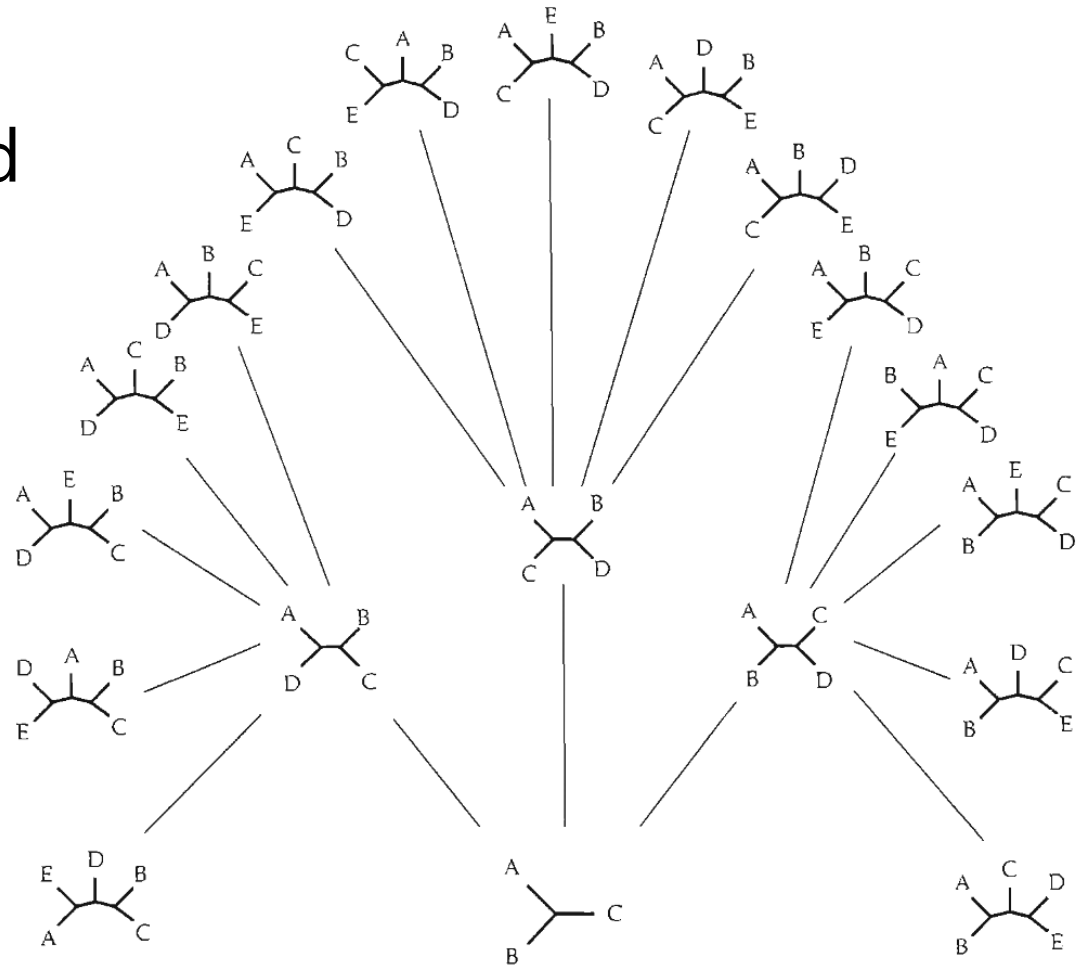


Figure 5.3: Search tree for most parsimonious tree in a five-species case.

Branch and bound

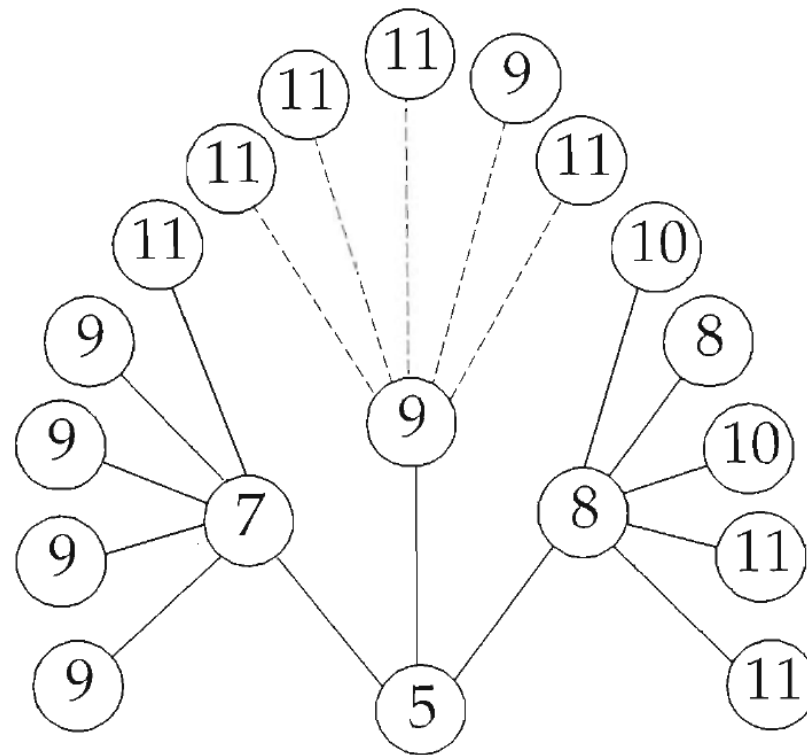


Figure 5.4: Search tree for most parsimonious tree for five species, using the data of Table 1.1. Trees are shown in Figure 5.3. Dashed lines are those not traversed by a branch and bound method. The species names in the data set correspond to labels A through E in Figure 5.3.

Felsenstein and the birth of statistical phylogenetics

How do we think about probabilities and trees?

Joe Felsenstein



Joe Felsenstein

Professor Emeritus of Genome Sciences, and of Biology, [University of Washington, Seattle](#)

Verified email at uw.edu - [Homepage](#)

[Evolutionary biology](#) [phylogenetic methods](#) [population genetics](#)

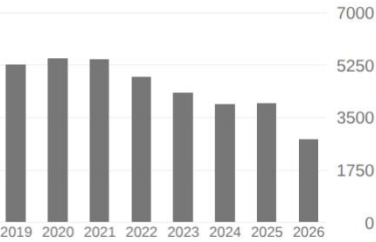
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PHYLIP (phylogeny inference package), version 3.5 c J Felsenstein Joseph Felsenstein.	32018 *	1993
Evolutionary trees from DNA sequences: a maximum likelihood approach J Felsenstein Journal of molecular evolution 17 (6), 368-376	17696	1981
Phylogenies and the comparative method J Felsenstein The American Naturalist 125 (1), 1-15	11884	1985
Inferring phylogenies J Felsenstein Inferring phylogenies, 664-664	6645	2004
Cases in which parsimony or compatibility methods will be positively misleading J Felsenstein Systematic zoology, 401-410	4387	1978

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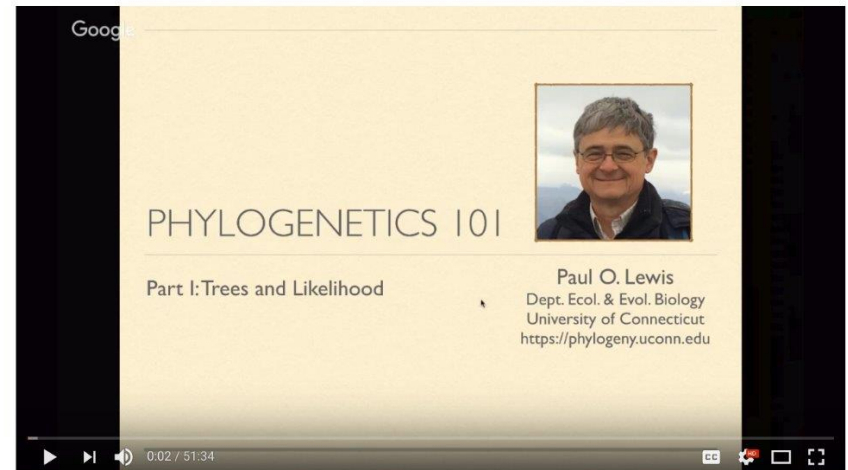
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phyloseminar.org
Streamed live on Apr 18, 2018

SUBSCRIBED 973



Primer part 1: tree terminology and substitution models

Slides: <https://git.io/vplW9>

SHOW MORE

If two DNA sequences are unrelated, what percent of bases (aligned sites) do you expect to be identical?

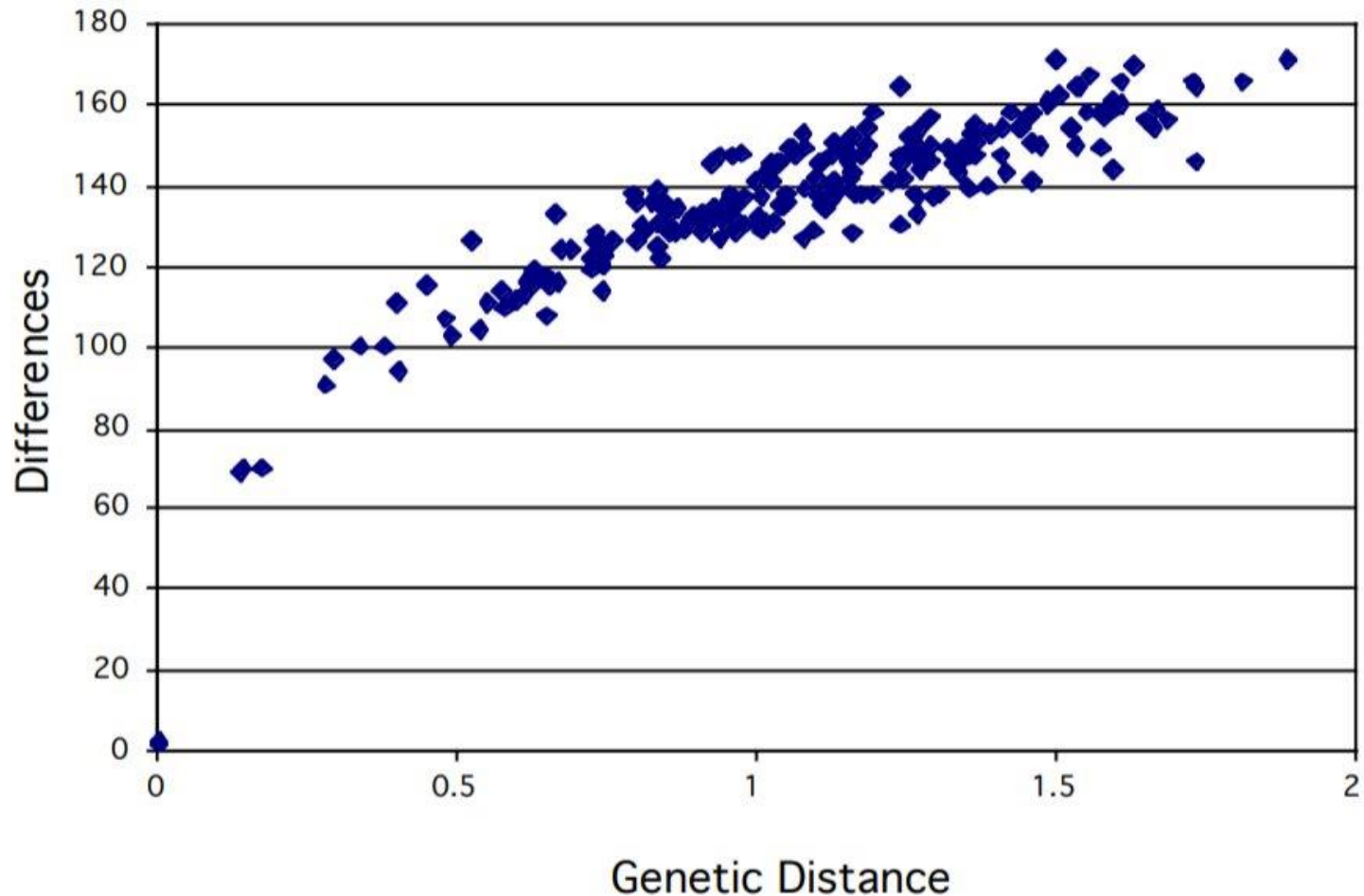
A. 50%

B. 25%

C. 0%

D. I need more information

Why do we need statistics and probability?



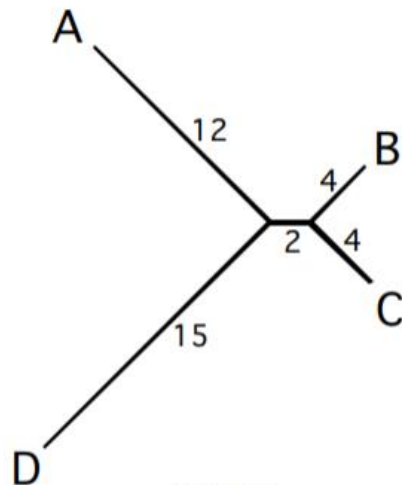
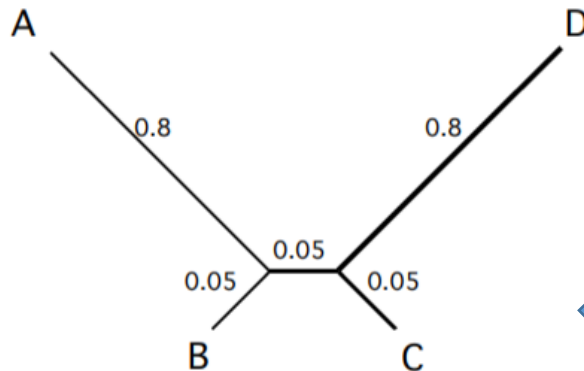
Long branch attraction

CASES IN WHICH PARSIMONY OR COMPATIBILITY METHODS WILL BE POSITIVELY MISLEADING¹

JOSEPH FELSENSTEIN

Abstract

Felsenstein, J. (Department of Genetics, University of Washington, Seattle, WA 98195) 1978. *Cases in which parsimony or compatibility methods will be positively misleading*. *Syst. Zool.* 27:401-410.—For some simple three- and four-species cases involving a character with two states, it is determined under what conditions several methods of phylogenetic inference will fail to converge to the true phylogeny as more and more data are accumulated. The methods are the Camin-Sokal parsimony method, the compatibility method, and Farris's unrooted Wagner tree parsimony method. In all cases the conditions for this failure (which is the failure to be statistically consistent) are essentially that parallel changes exceed informative, nonparallel changes. It is possible for these methods to be inconsistent even when change is improbable a priori, provided that evolutionary rates in different lineages are sufficiently unequal. It is by extension of this approach that we may provide a sound methodology for evaluating methods of phylogenetic inference. [Numerical cladistics; phylogenetic inference; maximum likelihood estimation; parsimony; compatibility.]



MP Tree

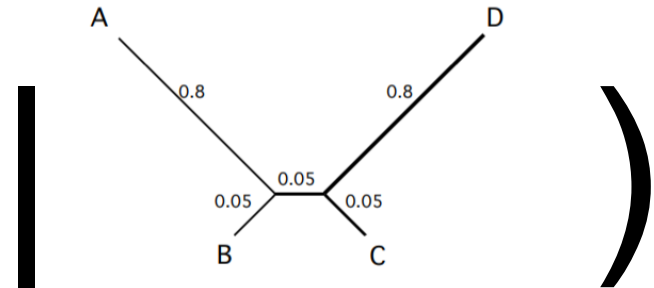
A
B
C
D

ATCGAGCAGCCTGGGAGAGAGACTTATTTGACAAACGTAA
ATTGGGGAGTAGCGTAAACACTCTTATTTGACGAAATTAT
ATCGTGGGTTAGAGTAGAGACTCTCATTTGACGAAATTAT
AACGTGGCGAATAGTAGTCAAAAAATGTGTACCAGATTAC

Conditional probabilities

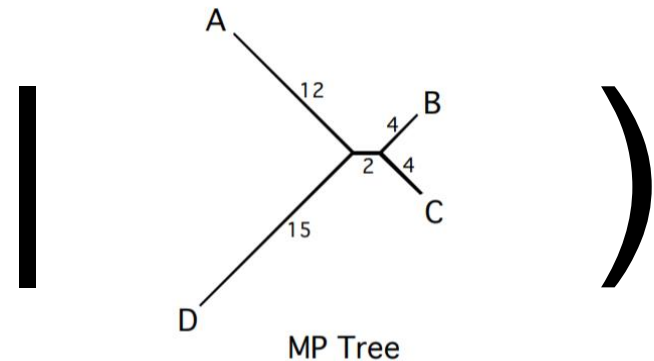
P(

```
ATCGAGCAGCCTGGGAGAGAGACTTATTTGACAAACGTAA
ATTGGGGAGTAGCGTAAACACTCTTATTTGACGAAATTAT
ATCGTGGGTTAGAGTAGAGACTCTCATTGACGAAATTAT
AACGTGGCGAATAGTAGTCAAAAAATGTGTACCAGATTAC
```



P(

```
ATCGAGCAGCCTGGGAGAGAGACTTATTTGACAAACGTAA
ATTGGGGAGTAGCGTAAACACTCTTATTTGACGAAATTAT
ATCGTGGGTTAGAGTAGAGACTCTCATTGACGAAATTAT
AACGTGGCGAATAGTAGTCAAAAAATGTGTACCAGATTAC
```



Read as: The probability of this alignment (the data) GIVEN or CONDITIONAL UPON this being the true tree.

How do we treat trees probabilistically?

Let's simplify to a binary (0/1) trait
(rather than 4 DNA bases)



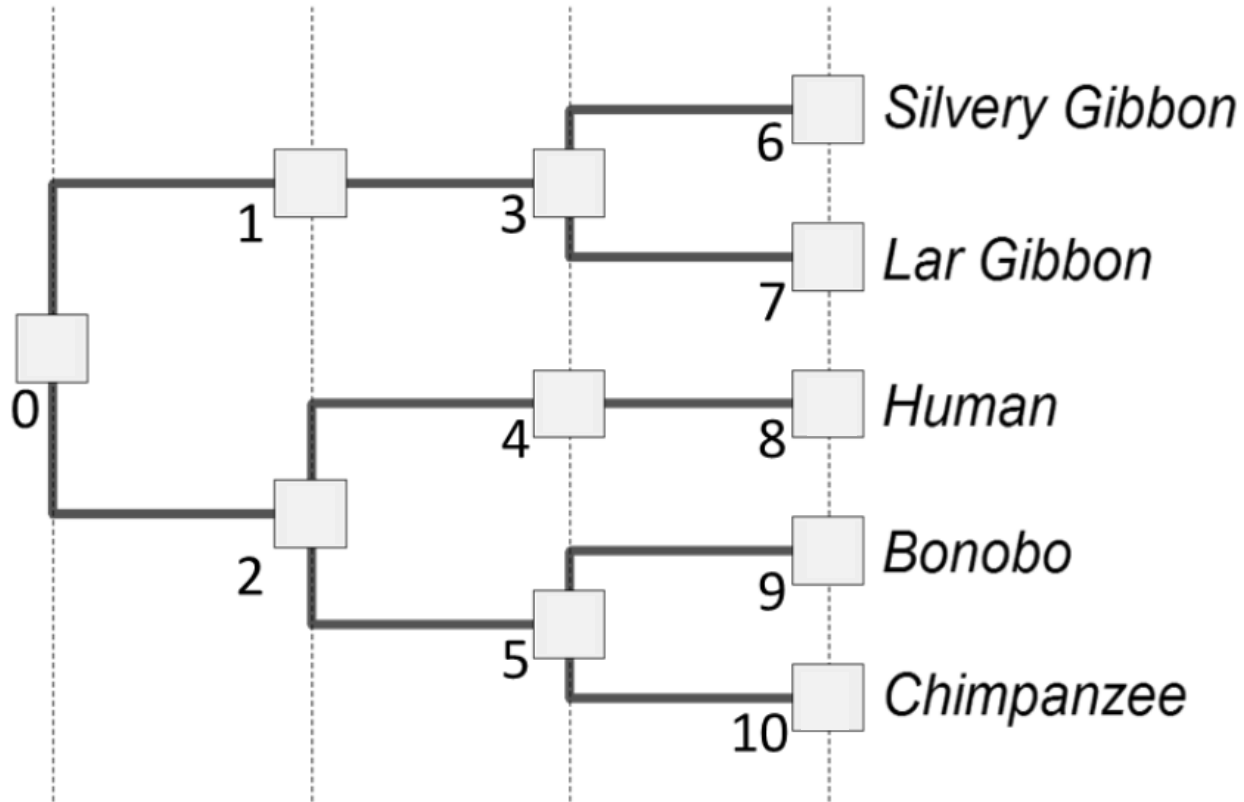
If 1 or 2 come up,
a substitution
occurs

Otherwise, no
substitution occurred

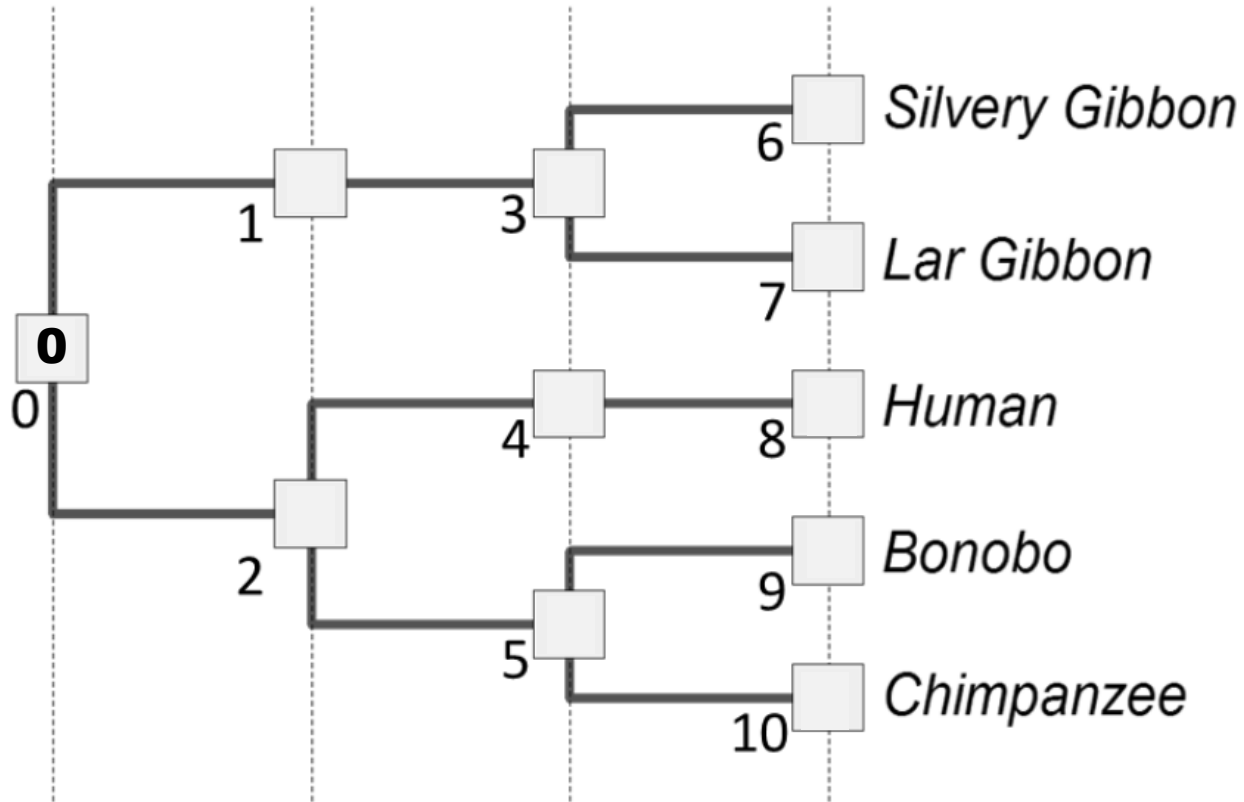


If a substitution occurs,
choose *which* substitution
occurs by a coin flip.

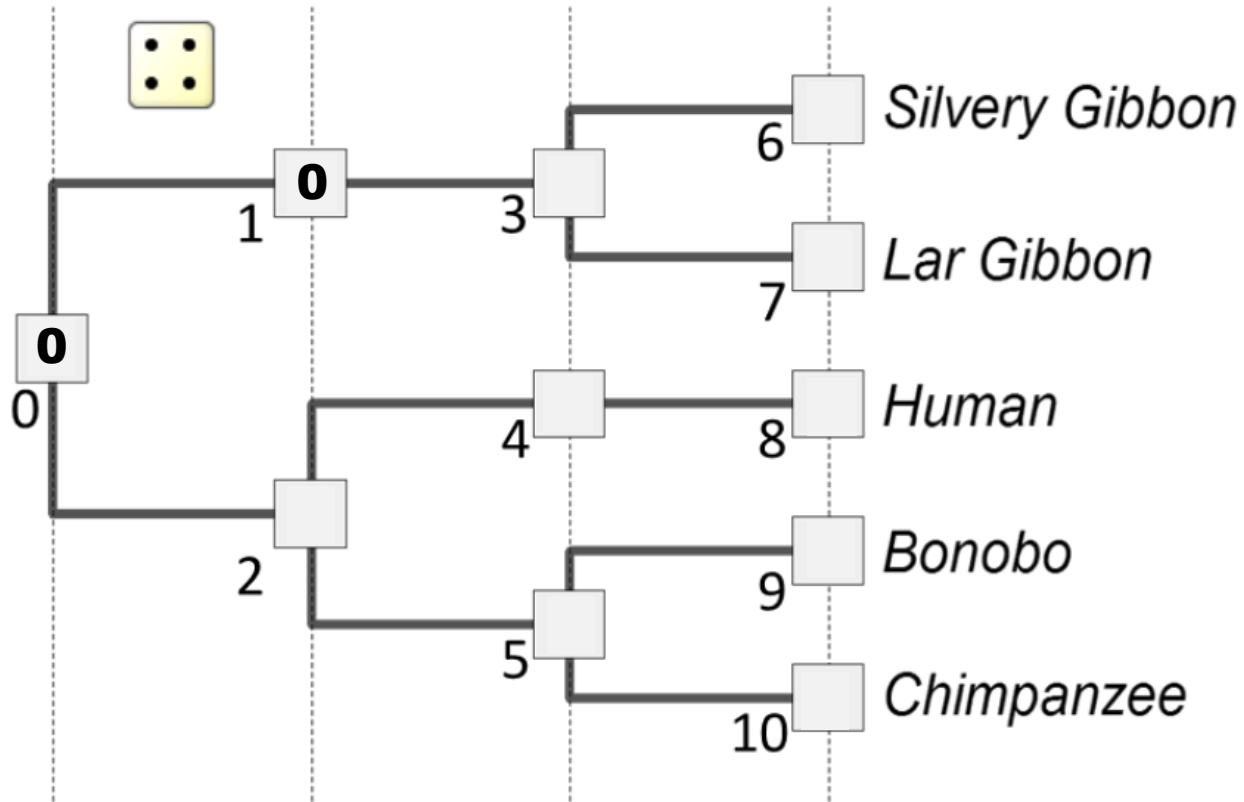
Generating data on a tree



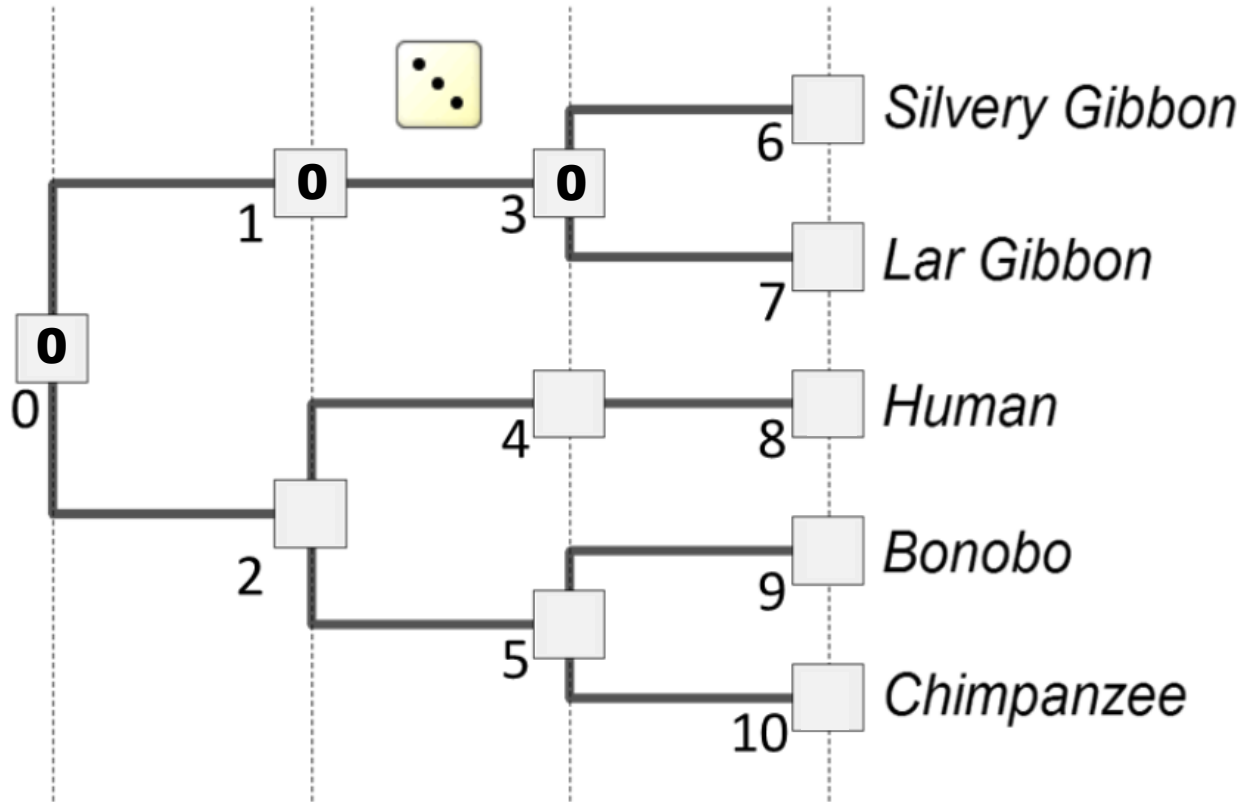
Initialize root



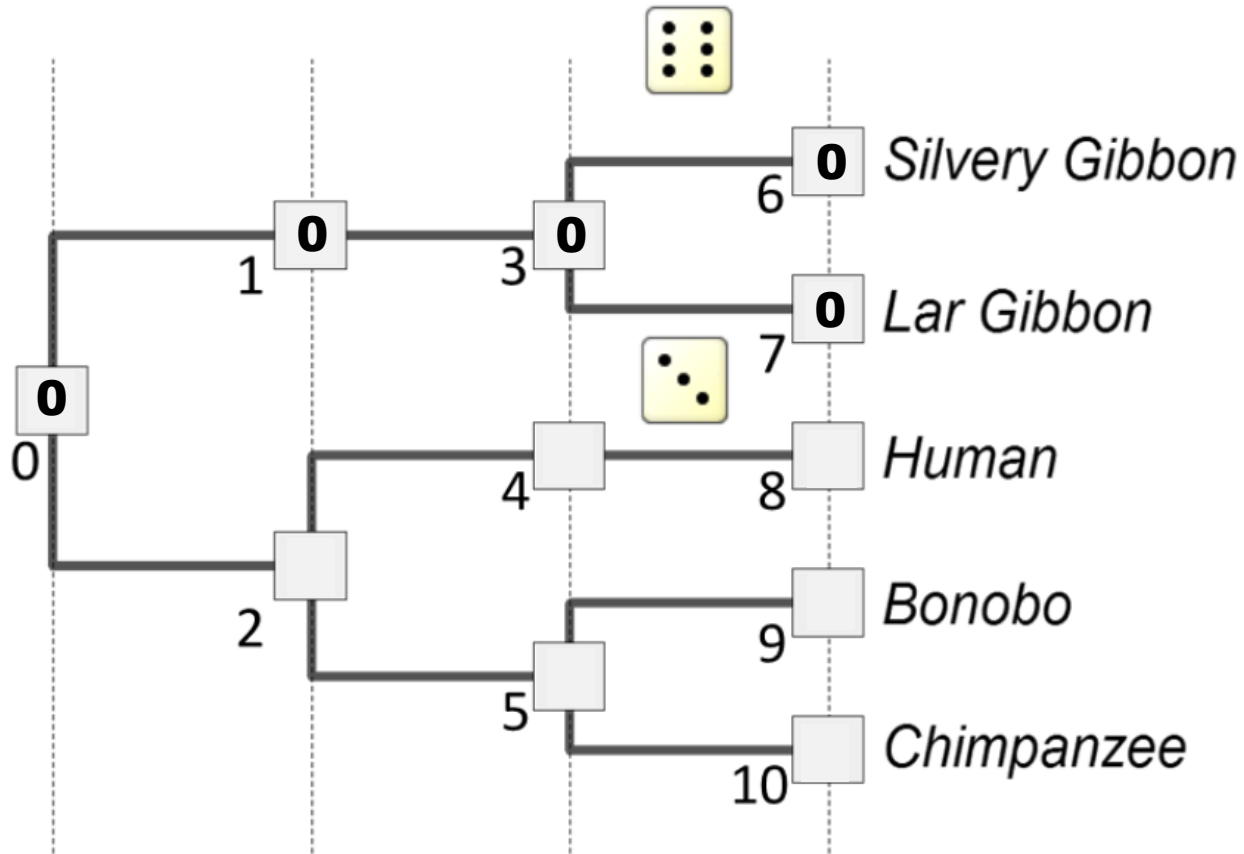
Step 1



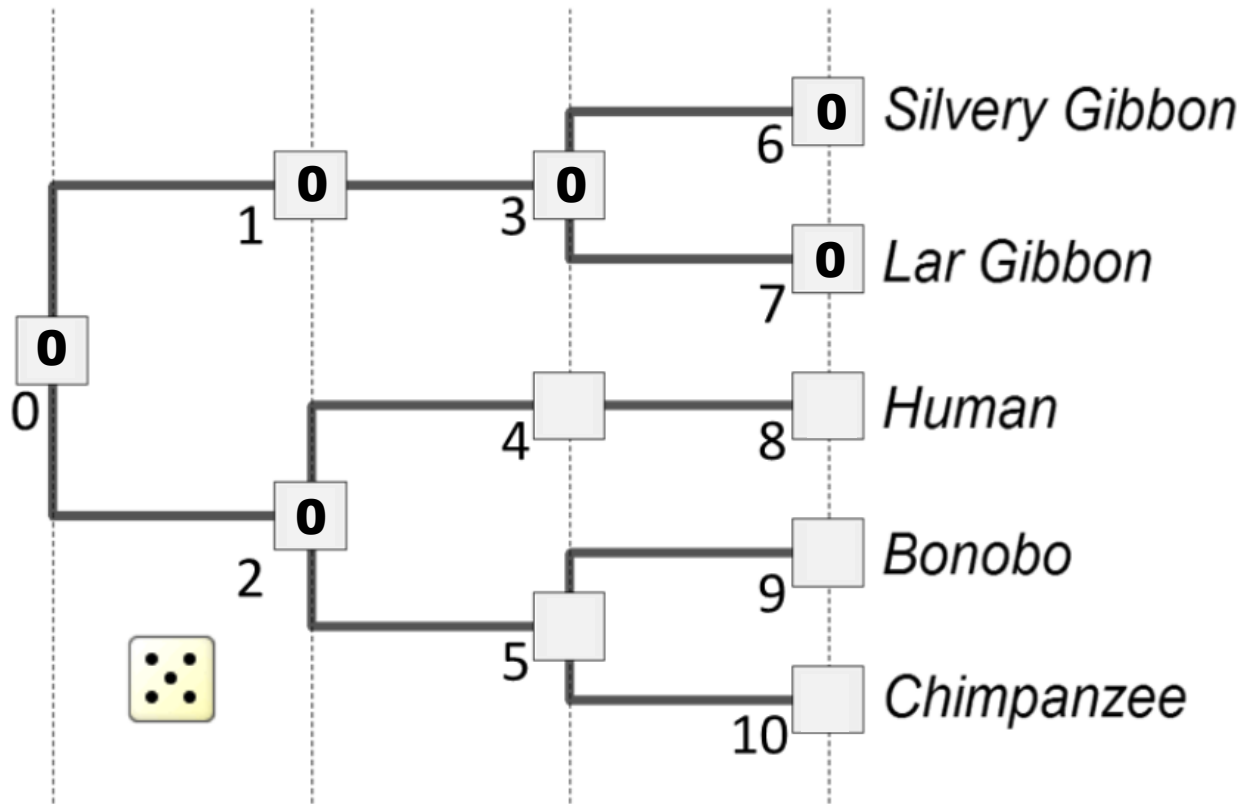
Step 2



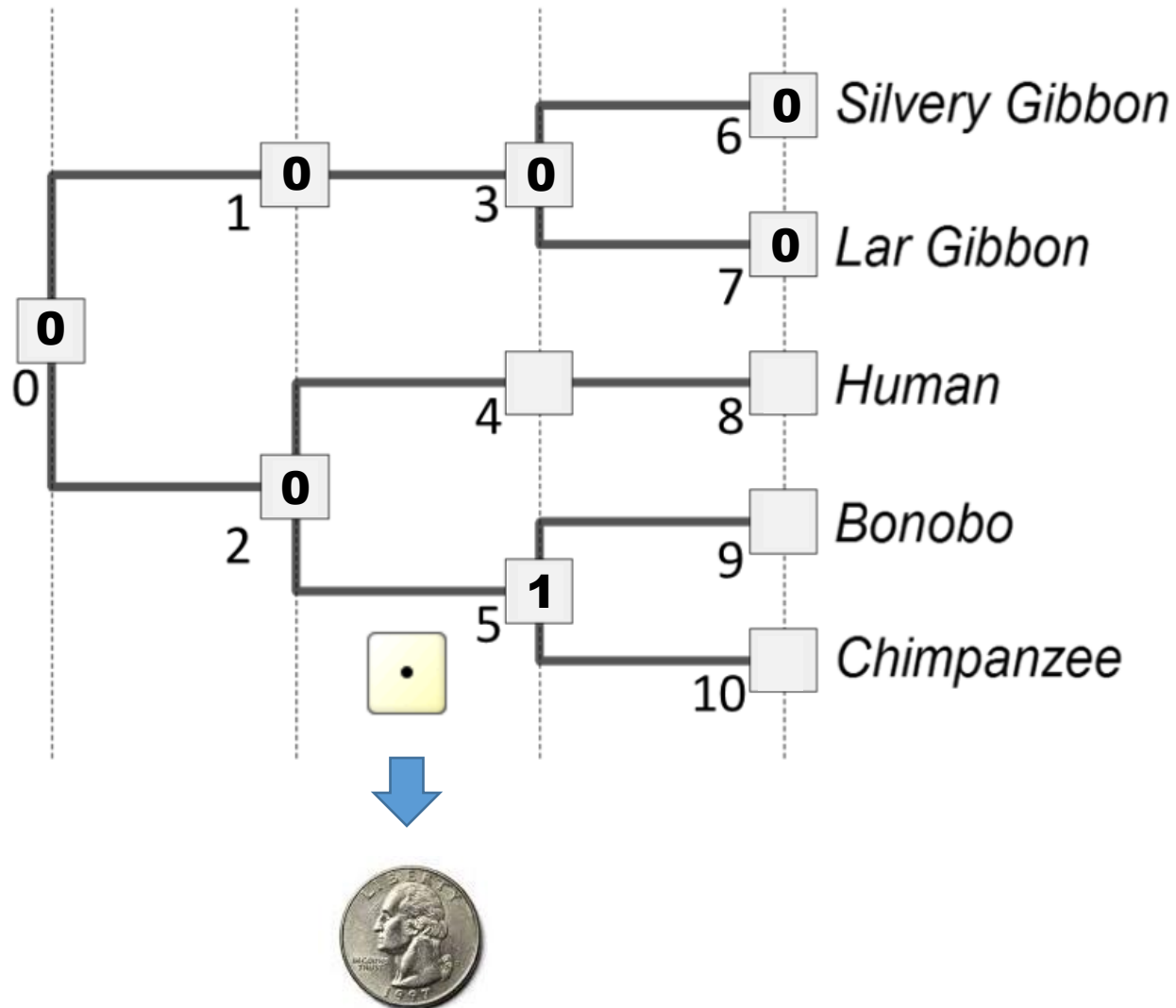
Step 3 and 4



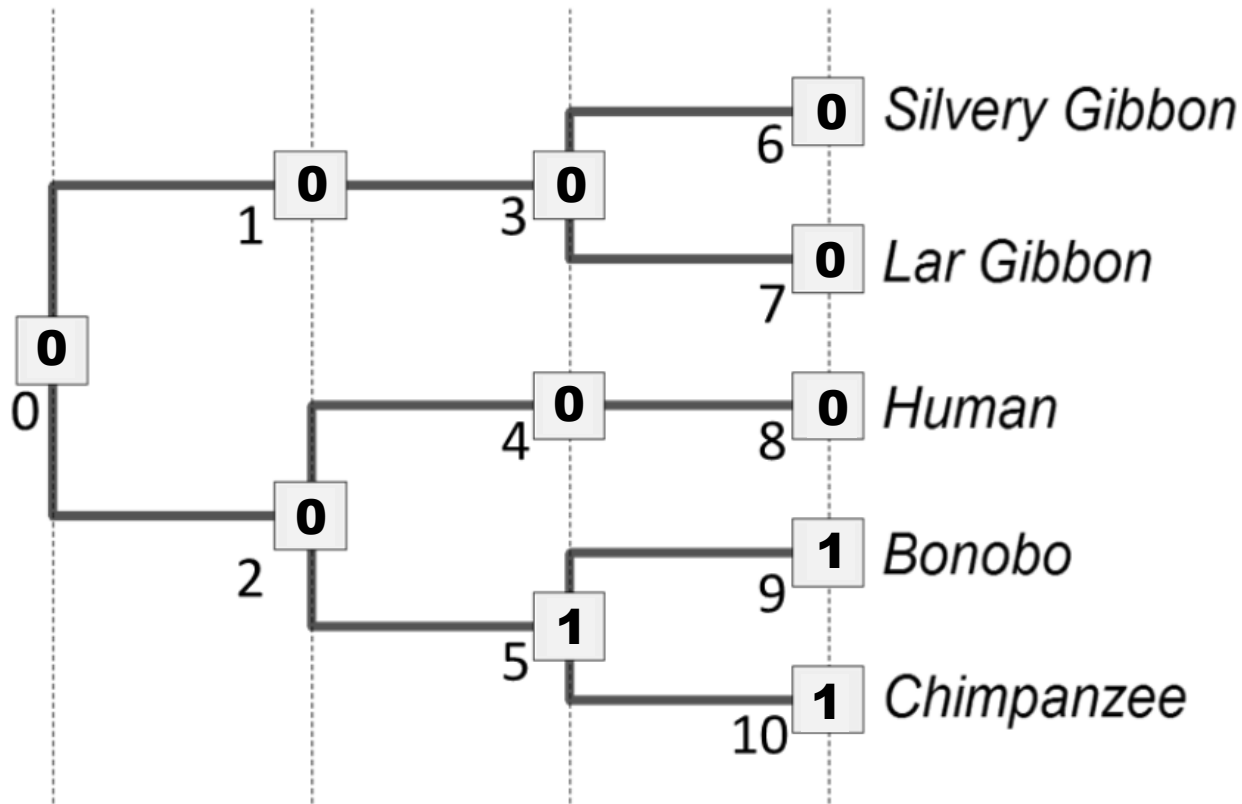
Step 5



Step 6



Step 7-10



What was the probability of this outcome?

Combining probabilities: The AND rule

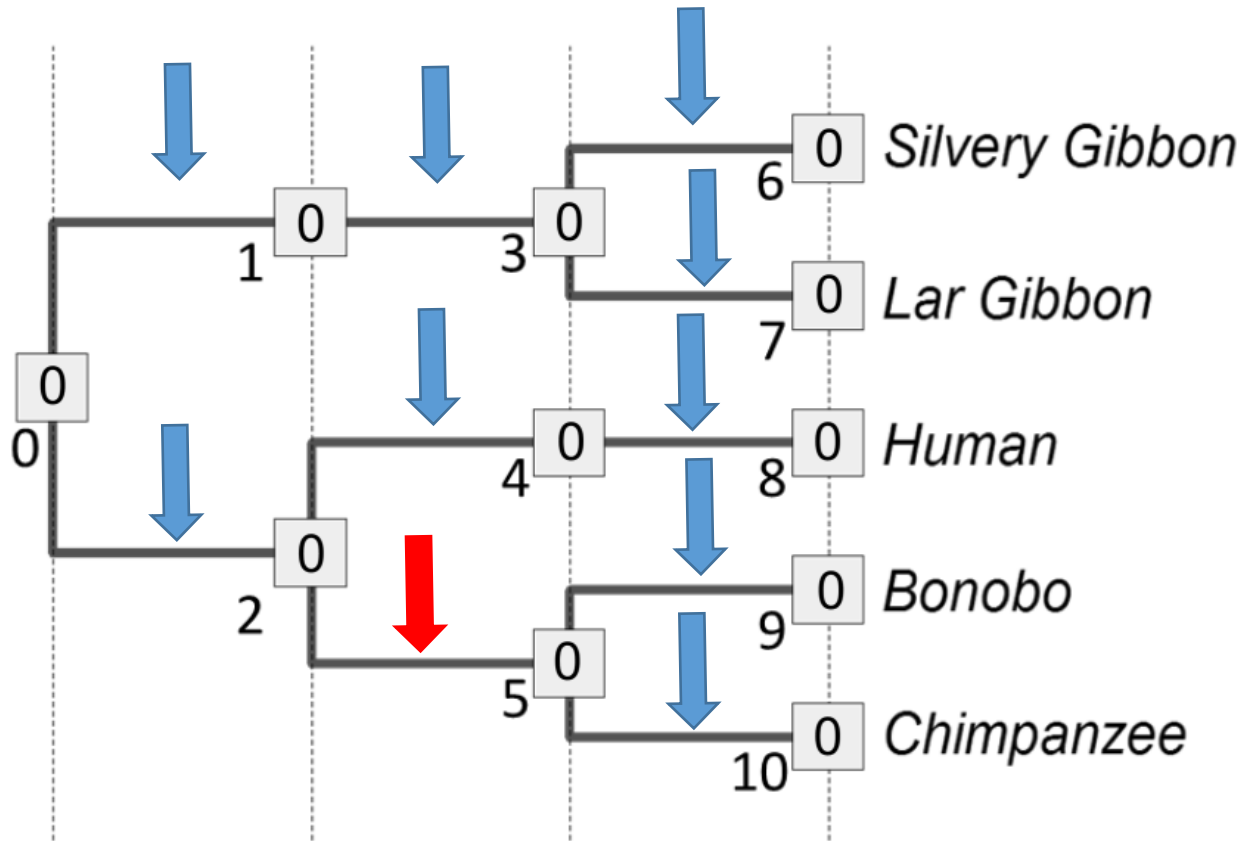
If two independent events occur, MULTIPLY their individual probabilities to get the full probability of an event. Each roll of the dice (and each flip of the coin) *is independent*.

Using 2 dice, what is the probability of



$$(1/6) \times (1/6) = 1/36$$

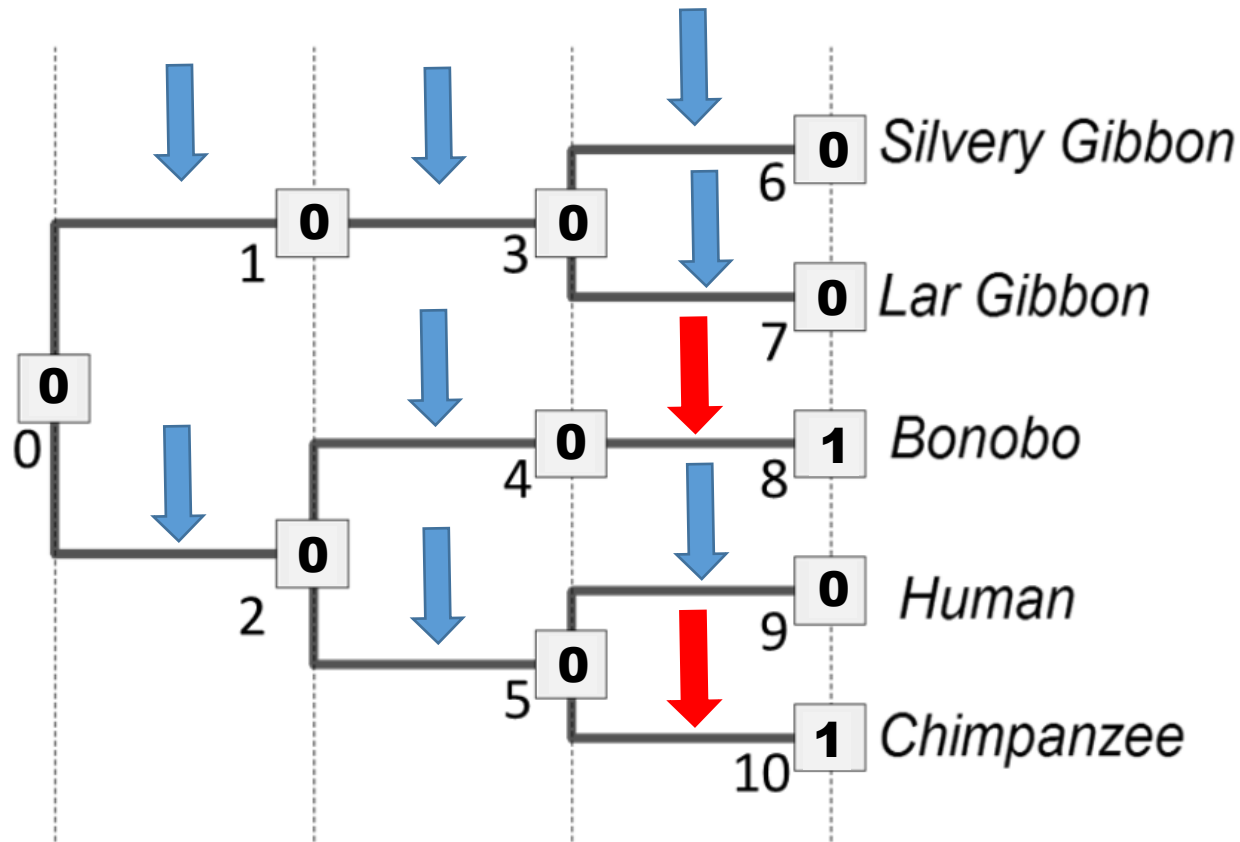
Combining probabilities: Calculating a likelihood



$$L(T_1 \mid D, N_i) = P(D, N_i \mid T_1)^* = (4/6)^9 \times [(2/6)(1/2)]^1 = 0.0043$$

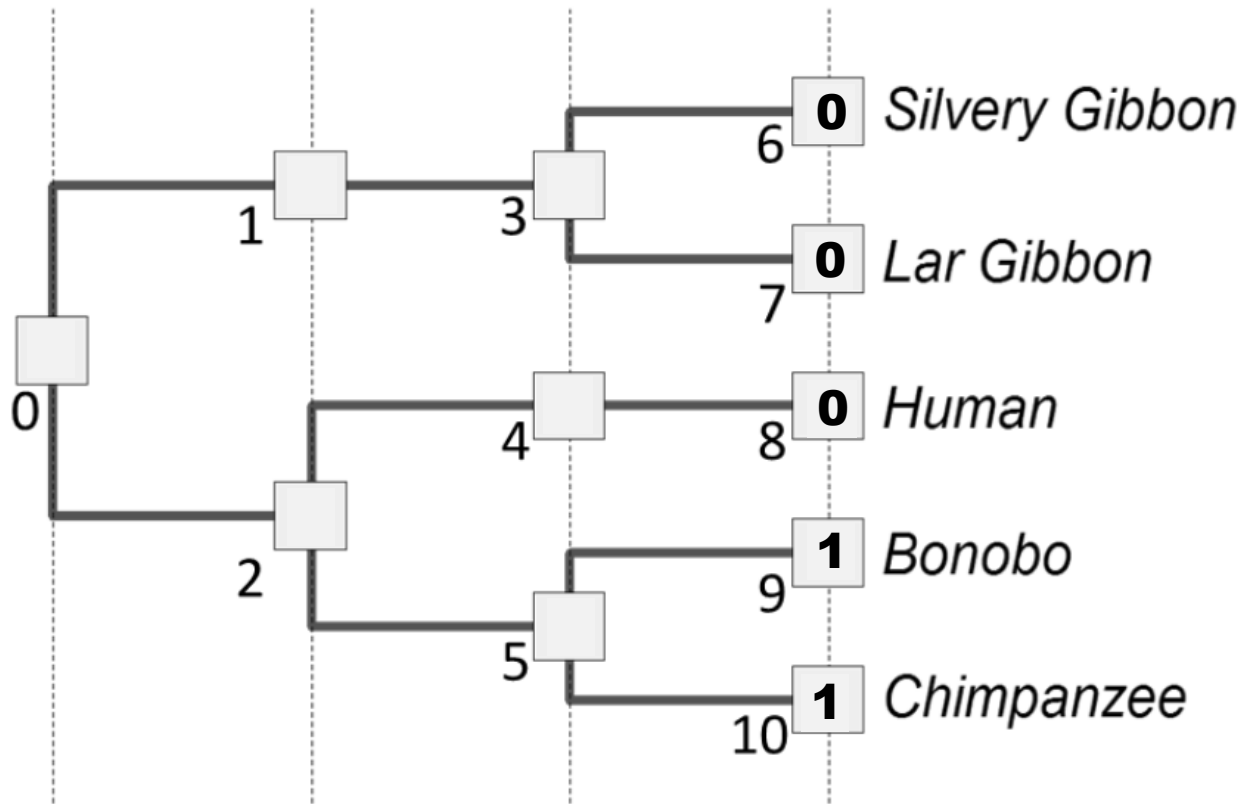
Read as: The likelihood of Tree 1 given the data at the tips (D) and nodes (N_i) is equal to the probability of the data and node states given Tree 1 is true. (We can't ask "is Tree 1 true" so simply, so we use the word Likelihood instead)

Combining probabilities: The Likelihood under an alternative topology



$$\text{Likelihood}(\mathbf{D}, \mathbf{N}_j | \mathbf{T}_2)^* = (4/6)^8 \times [(2/6)(1/2)]^2 = 0.0011$$

Node states are usually unknown



Combining probabilities: The OR rule

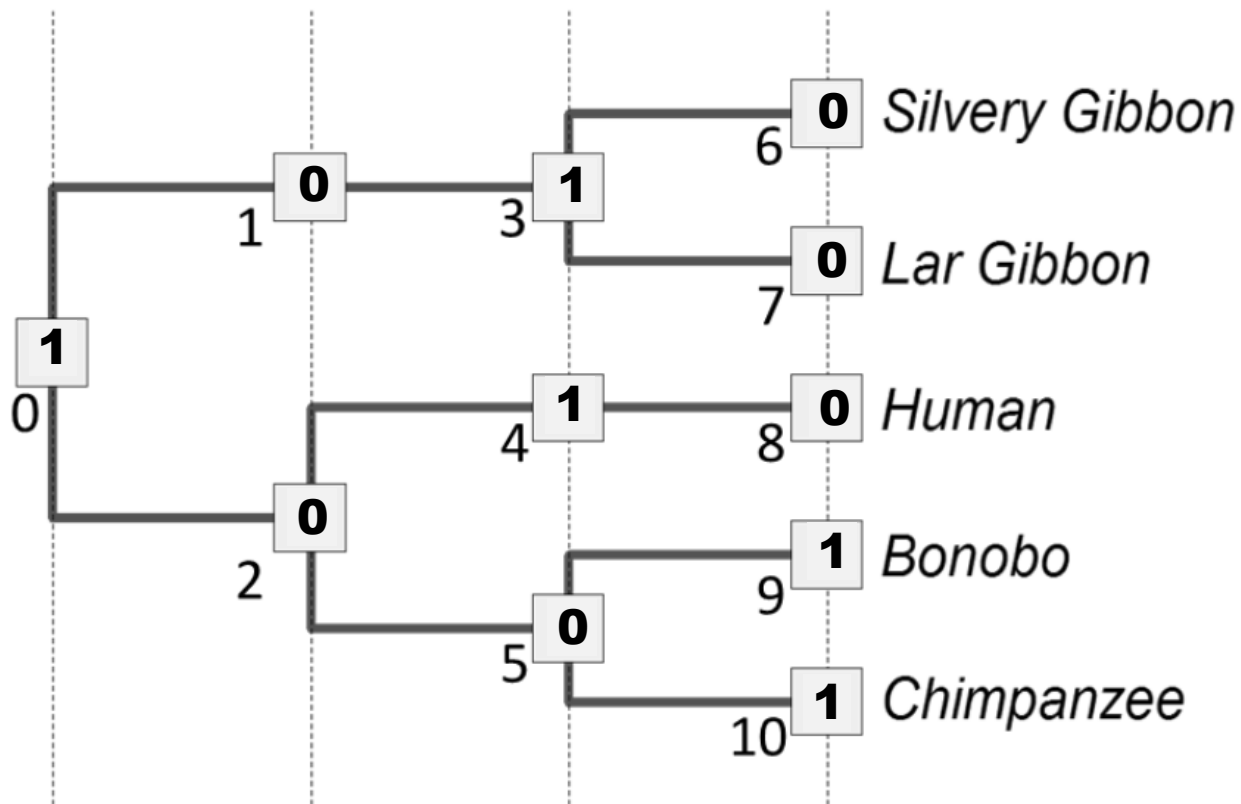
Two mutually exclusive probabilities should be ADDED together to get the total probability of the two events

Using one die, what is the probability of



$$(1/6) + (1/6) = 1/3$$

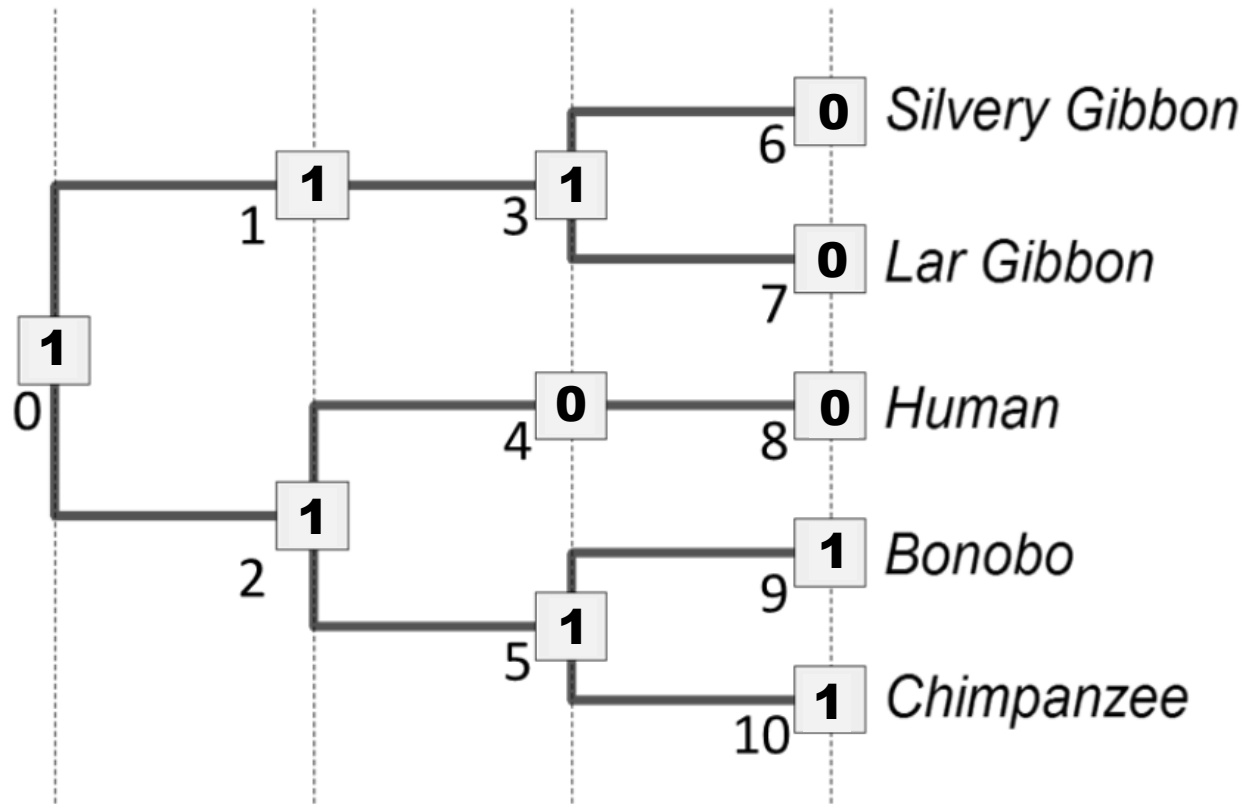
Combining probabilities:
Calculate likelihood of another possible history



Likelihood $(D, N_k | T_1) = (4/6)^1 \times [(2/6)(1/2)]^9 = 0.000000066$

This is a possible outcome under Tree₁, but not especially likely!

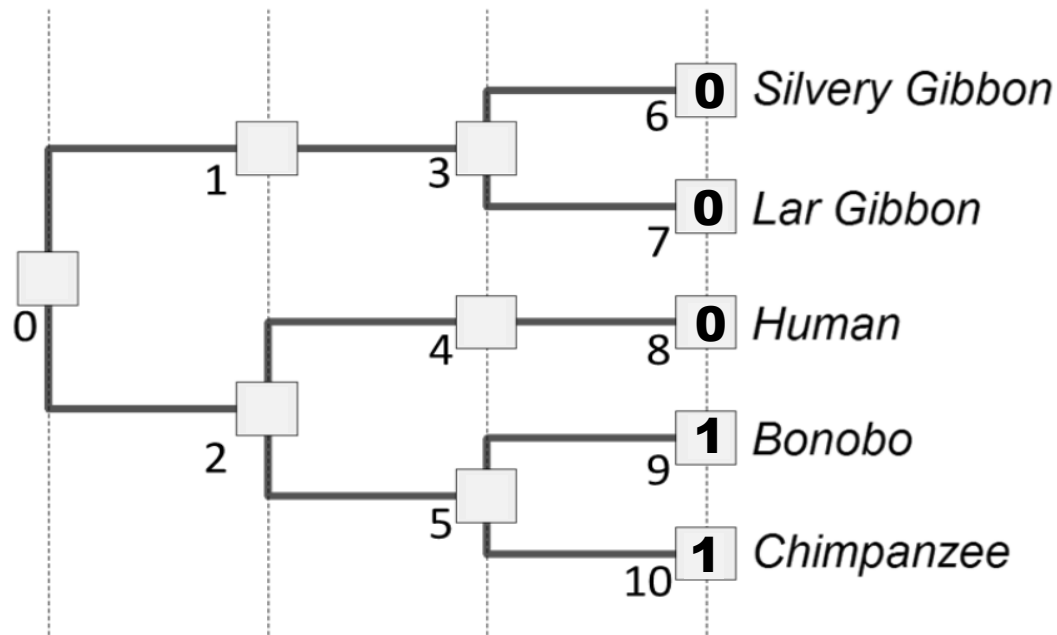
Combining probabilities: Yet another history under the same tree



$$\text{Likelihood } (D, N_k | T_1) = (4/6)^7 \times [(2/6)(1/2)]^3 = 0.00027$$

This is also a possible outcome, a little bit more likely!

Calculating likelihoods over all possible histories



There are $2^6 = 64$ possible node assignments for Tree_1 . We could calculate the likelihood of each one, then _____ them together, to get the total $\text{Likelihood}(\text{Tree}_1 | \text{Data})$.

Now, like parsimony, you can do either a heuristic or exhaustive search for the tree with the maximum likelihood)

Biology will want *continuous time*

We used discrete time blocks to do our dice rolling

Change in evolution can happen anytime, continuously

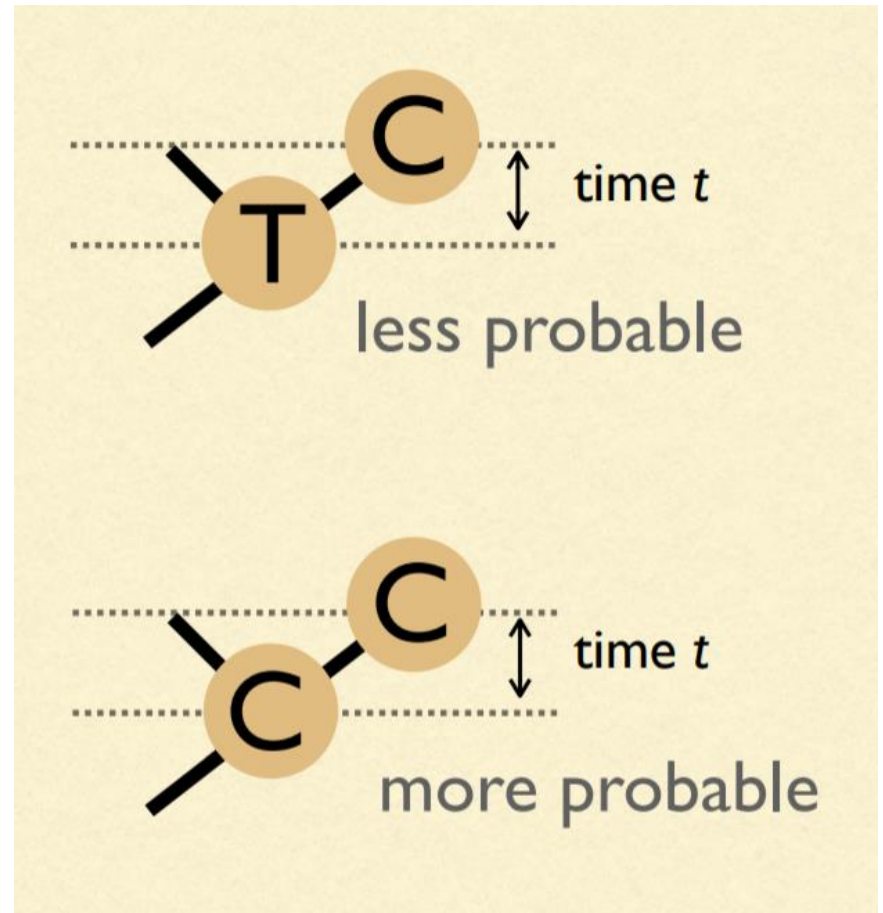
Continuous-time Markov Models

Markov assumption = probability of change depends only on the current state, not how long it has been in that state.

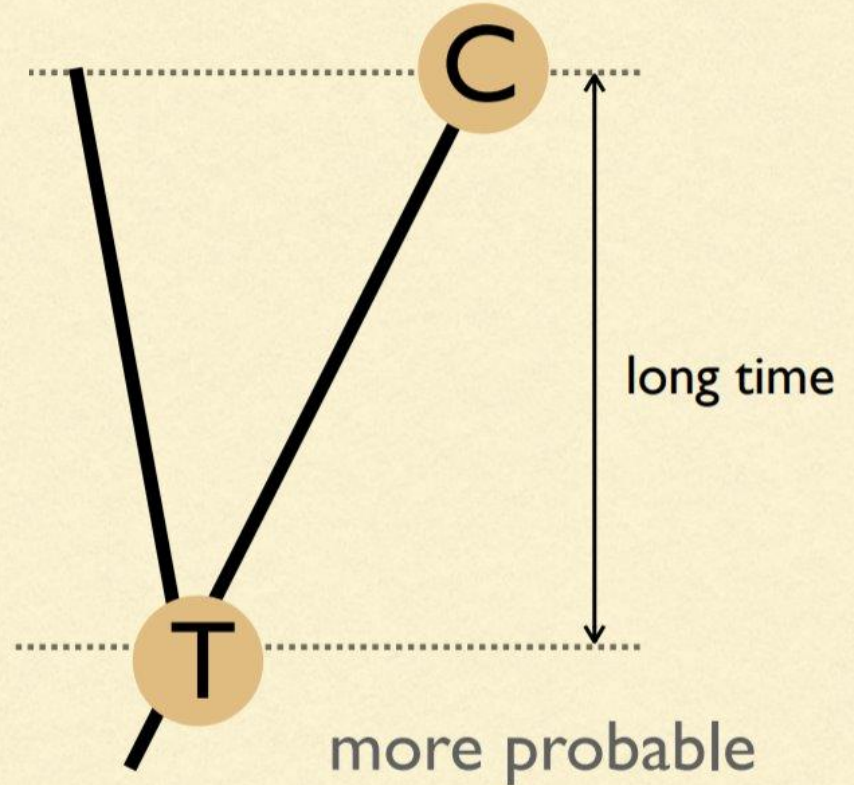
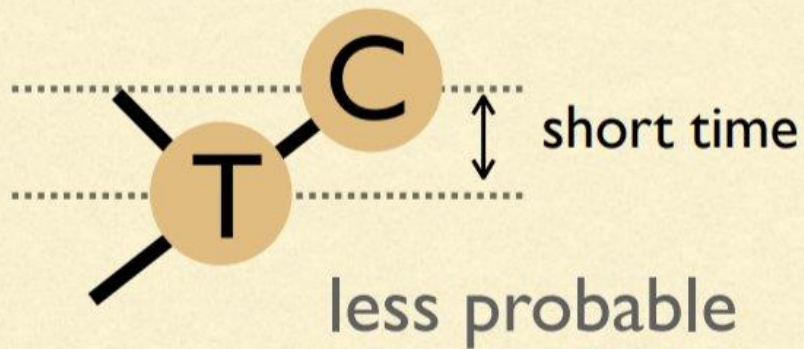
Our model of change depends on time: must have branch lengths

Units of branch length
will be expected
number of
substitutions per site

(= rate of substitutions x
time)

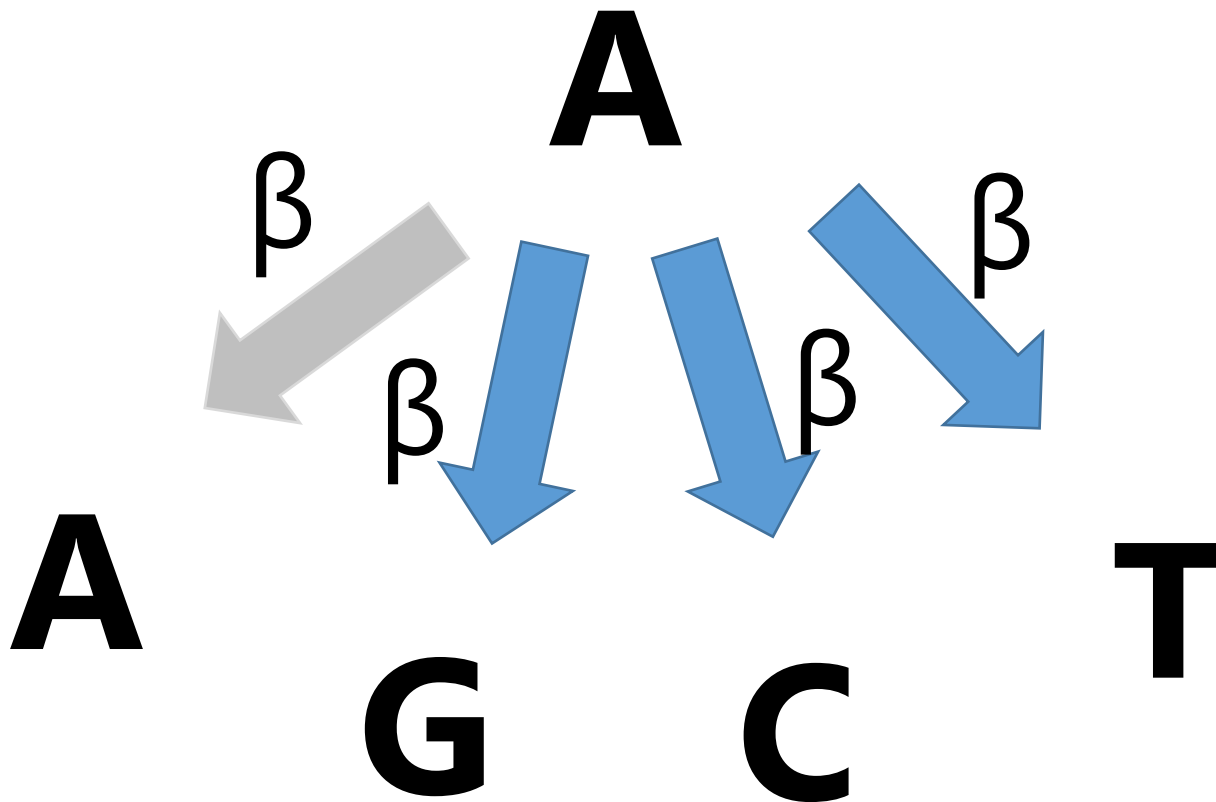


Probabilities are dependent on time



Event probabilities

$$\mu = 4\beta$$



CTMC on a (very) simple phylogeny...



$$P_{AA} =$$

Probability nothing happened +

Probability something happened, but that the last thing that happened ended in an A

Staying the same



$$P_{AA} = (e^{-\mu t}) + (1 - e^{-\mu t})(1/4)$$

Diagram illustrating the probability P_{AA} of staying in state A over time t .

The equation is broken down into two parts:

- $(e^{-\mu t})$: Probability something doesn't happen.
- $(1 - e^{-\mu t})(1/4)$: Probability at least one thing happens and the last thing that happened ends in an A.

A substitution



$$P_{AG} =$$

$$(1 - e^{-\mu t})(1/4)$$



Probability
at least
one thing
happens



Probability
that the last
thing that
happened
ends in an G

Another possible substitution



$$P_{AC} =$$

$$(1 - e^{-\mu t})(1/4)$$



Probability
at least
one thing
happens



Probability
that the last
thing that
happened
ends in an C

The last possible substitution



$$P_{AT} =$$

$$(1 - e^{-\mu t})(1/4)$$



Probability
at least
one thing
happens



Probability
that the last
thing that
happened
ends in an T

Substitutions vs. "events"

$$\nu = (3/4)\mu t = 3\beta t$$

$$4\nu/3 = \mu t$$

Only 3 out of 4 events results in a substitution. Thus, we can define the substitution rate ν .

Putting all four possible events together

$$P_{AA} = (e^{-\mu t}) + (1 - e^{-\mu t})(1/4)$$

$$P_{AG} = (1 - e^{-\mu t})(1/4)$$

$$P_{AC} = (1 - e^{-\mu t})(1/4)$$

$$P_{AT} = (1 - e^{-\mu t})(1/4)$$

Sanity check: do they all add to one?

$$P_{AA} = (1/4) + (3/4)(e^{-4\nu/3})$$

$$P_{AG} = (1/4) - (1/4)(e^{-4\nu/3})$$

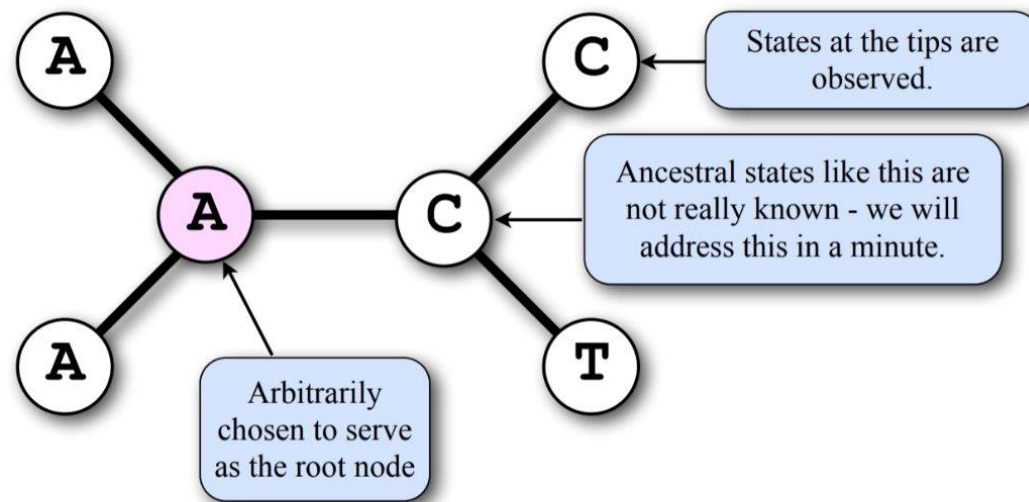
$$P_{AC} = (1/4) - (1/4)(e^{-4\nu/3})$$

$$P_{AT} = (1/4) - (1/4)(e^{-4\nu/3})$$

Likelihood of an unrooted tree

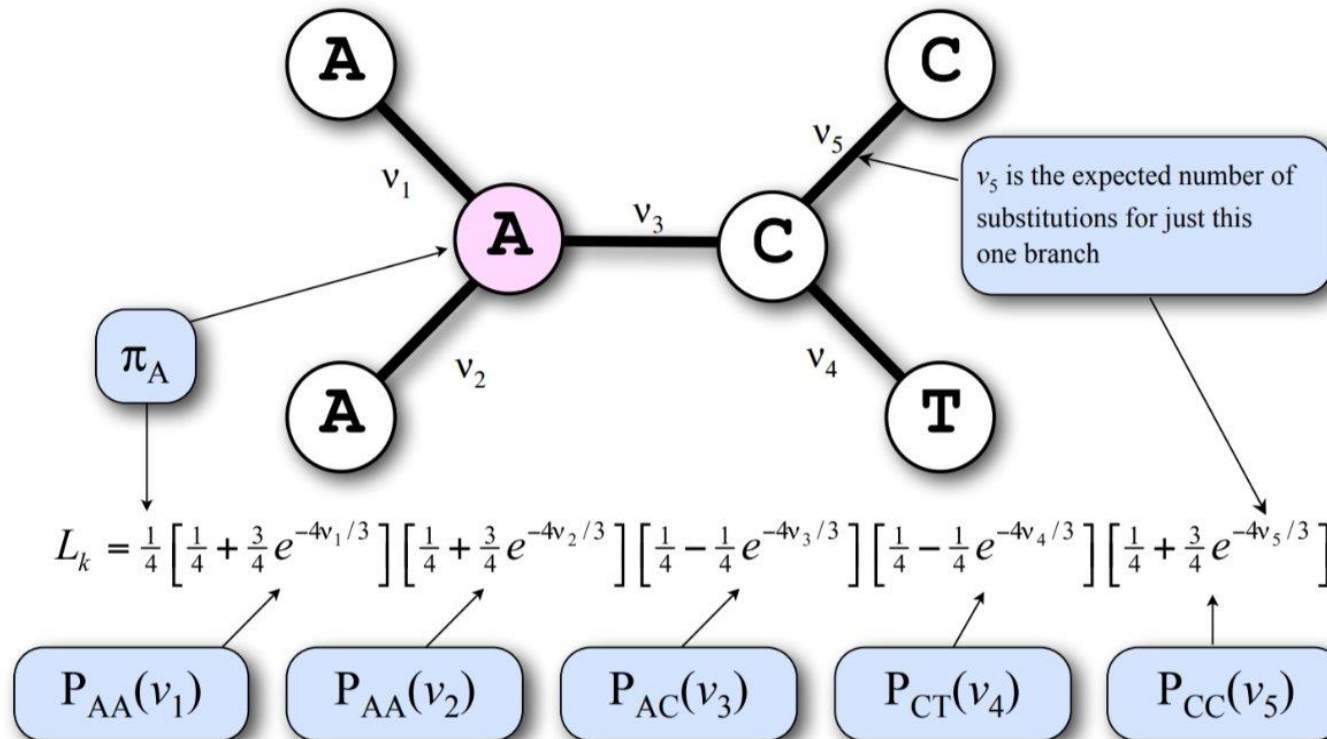
Likelihood of an unrooted tree

(data shown for only one site)



Calculating the likelihood of a single history and site

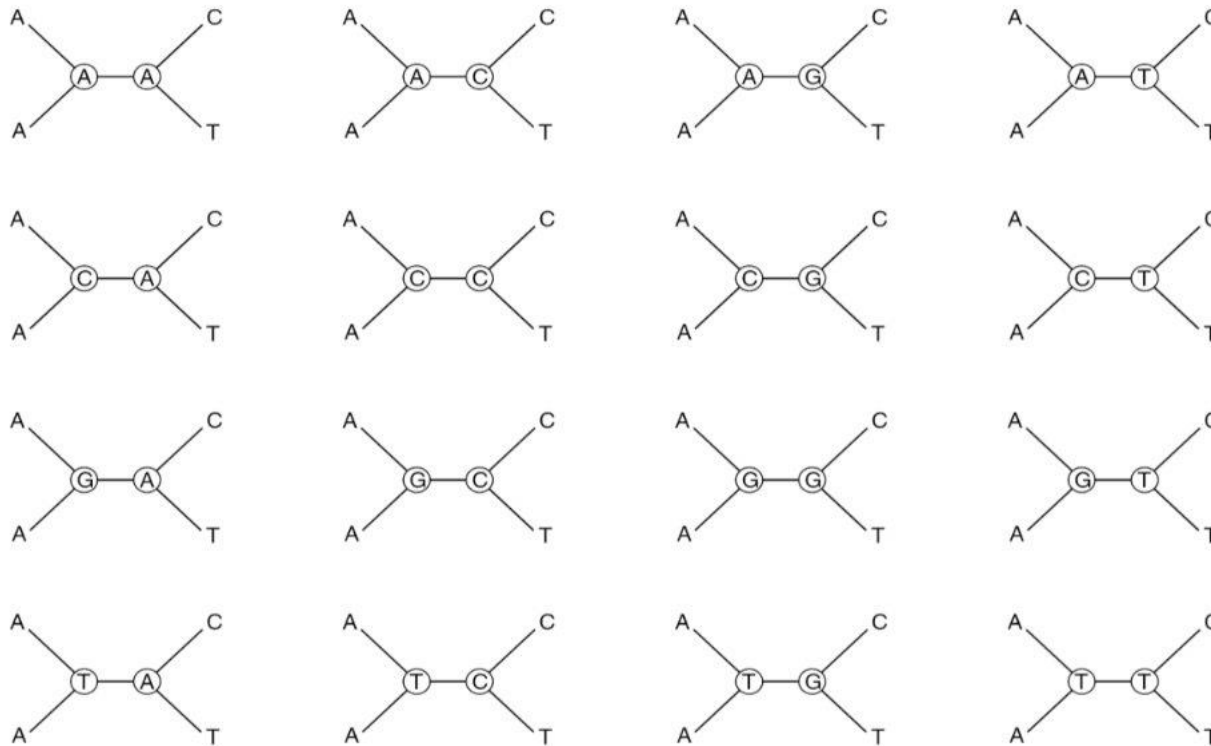
Likelihood for site k



Note use of the AND probability rule

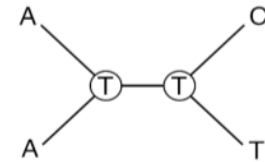
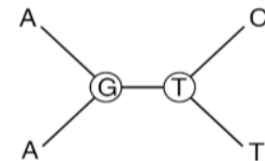
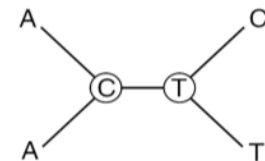
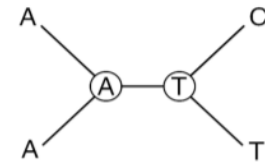
All possible node histories by brute force

Brute force approach would be to calculate L_k for all 16 combinations of ancestral states and sum them



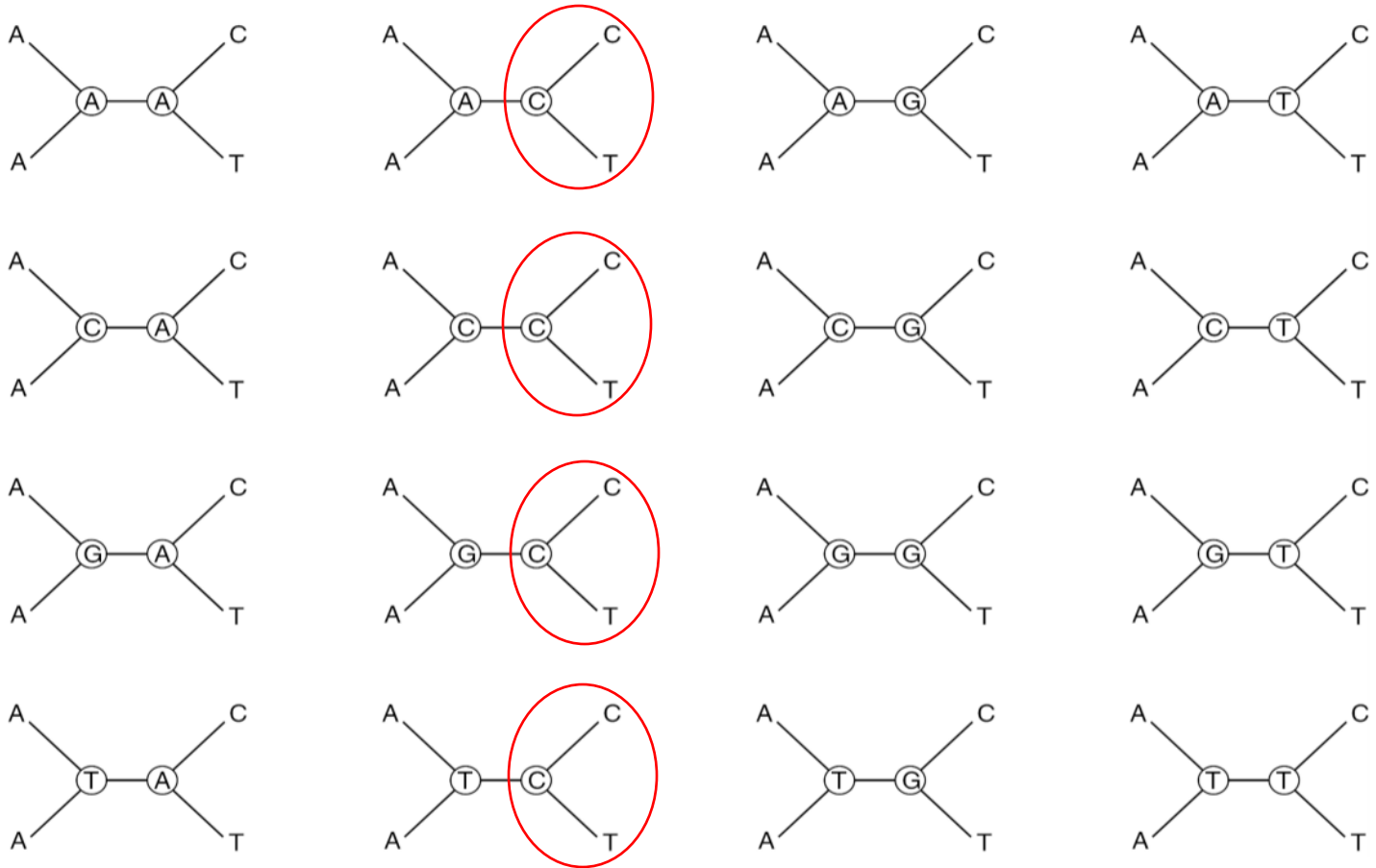
Note use of the OR probability rule

Repeated patterns in possible ancestries



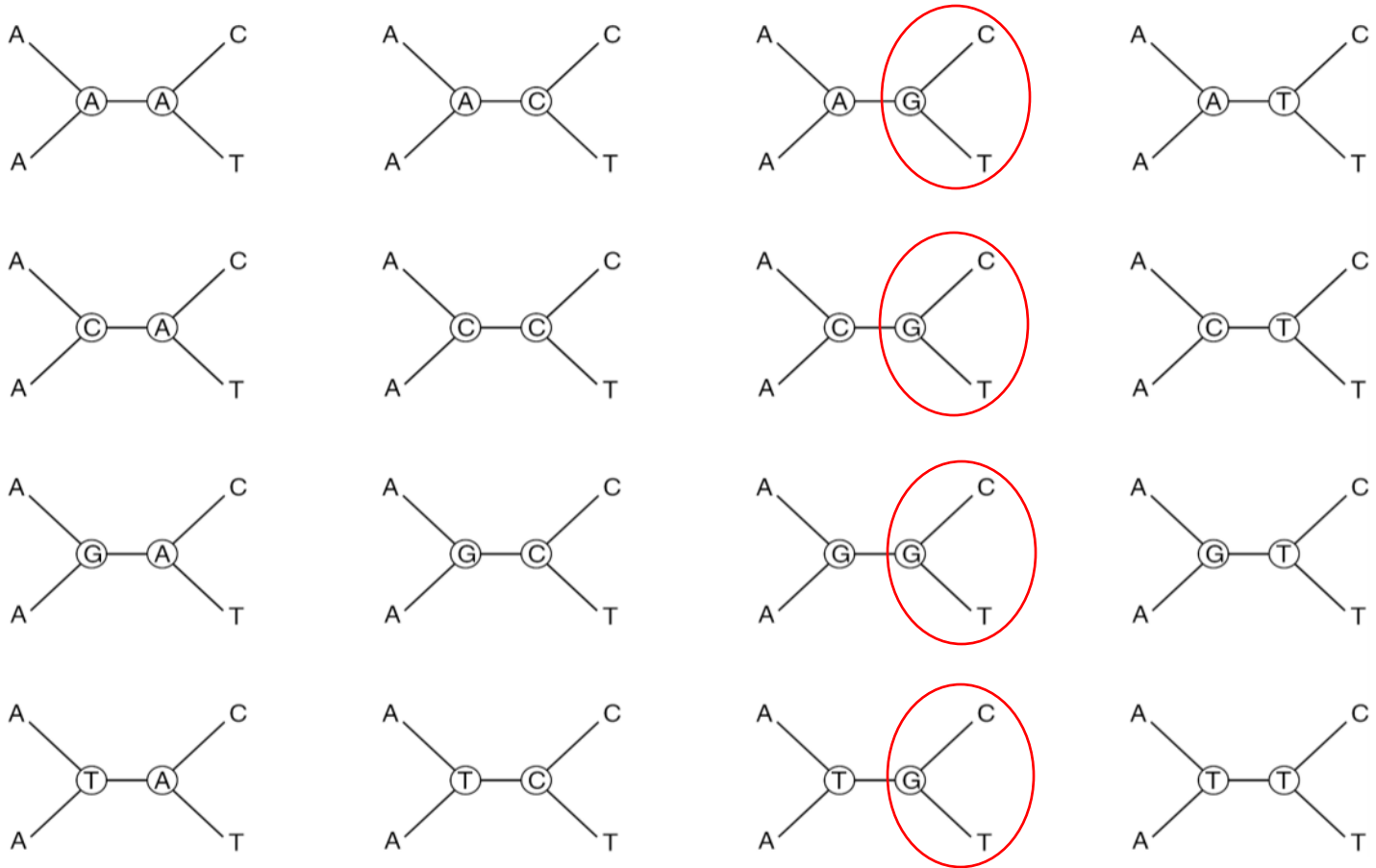
Note use of the OR probability rule

This repetition repeats for each set of 4 trees



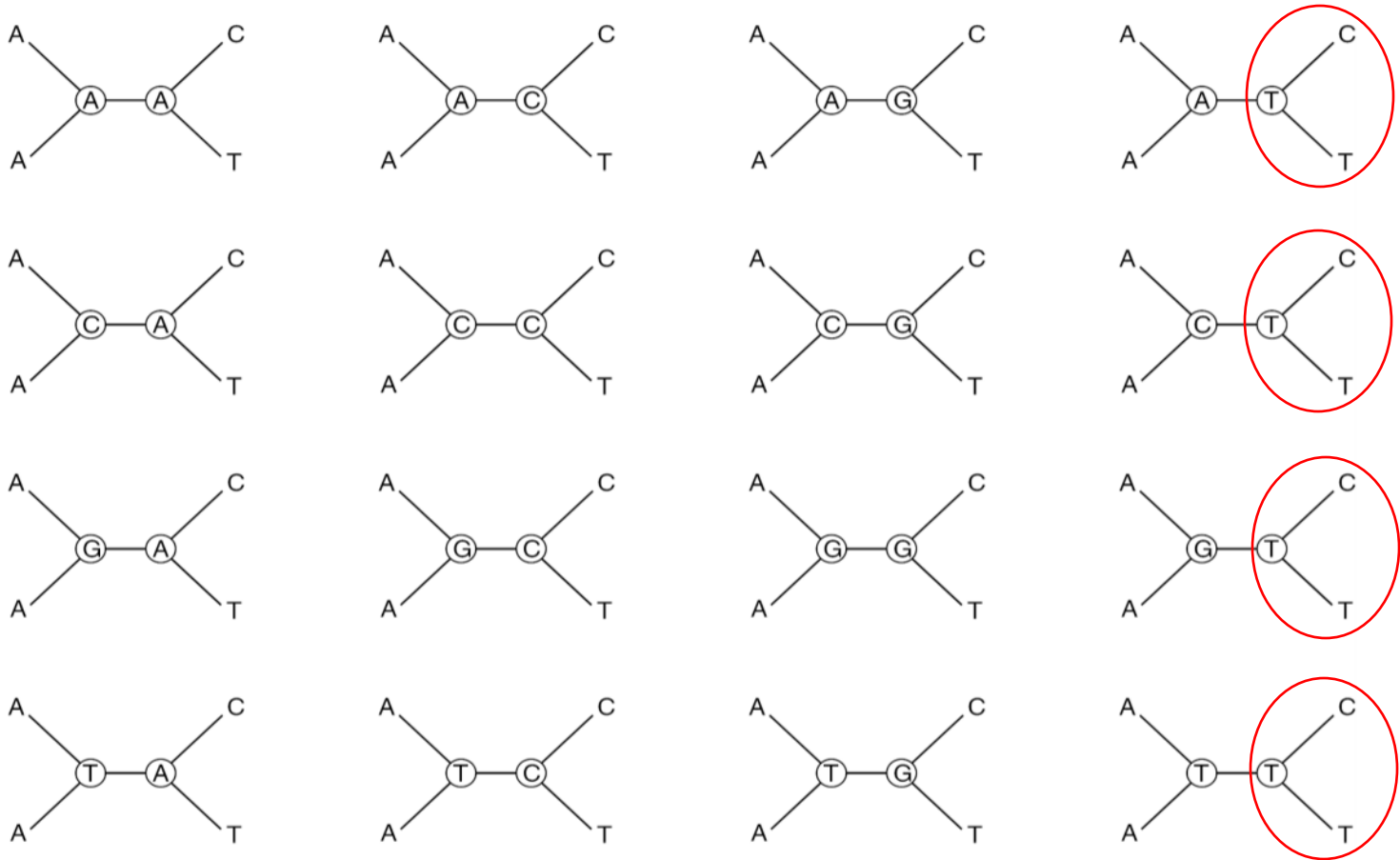
Note use of the OR probability rule

And it repeats again



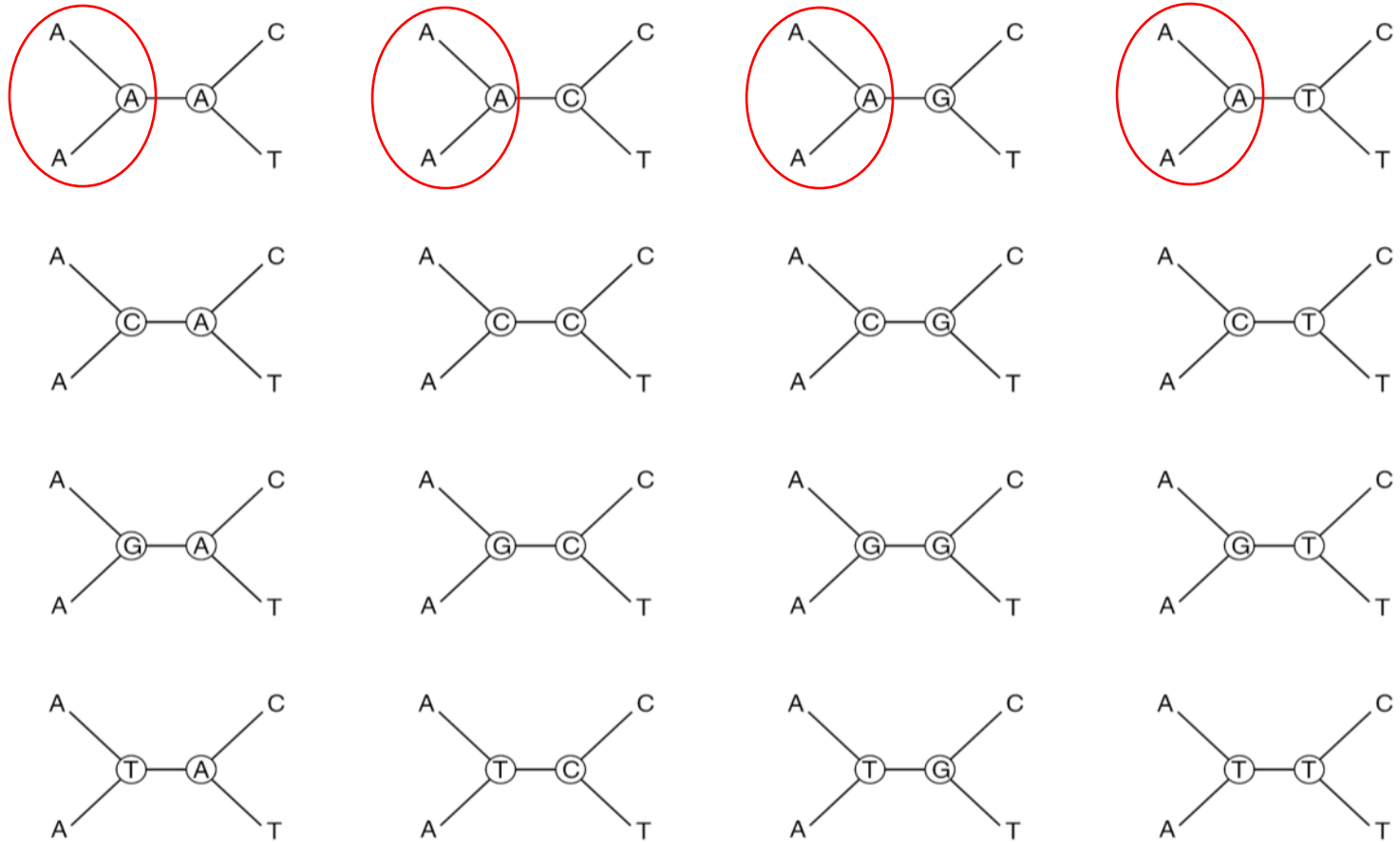
Note use of the OR probability rule

And for the last four



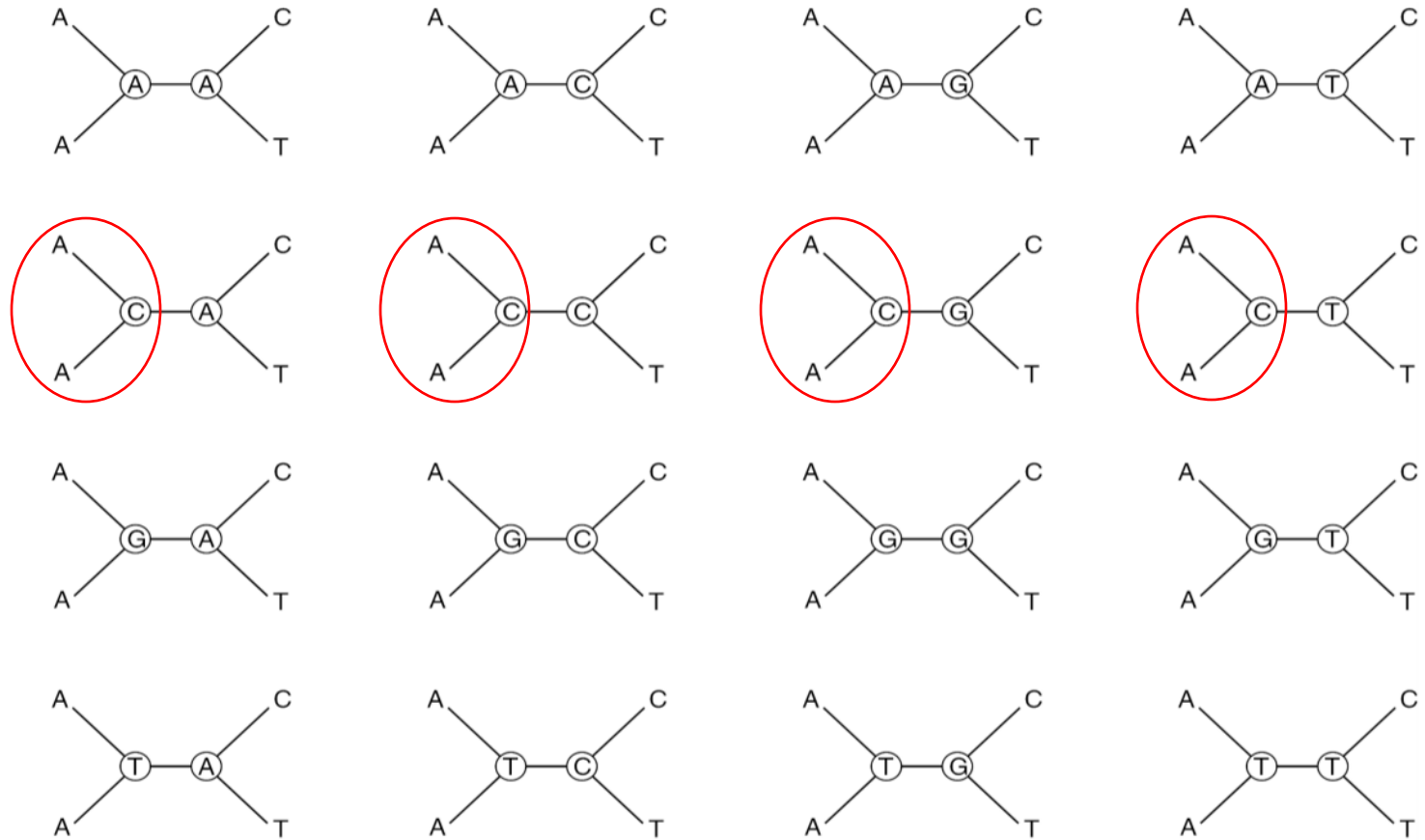
Note use of the OR probability rule

And shared bipartitions across rows



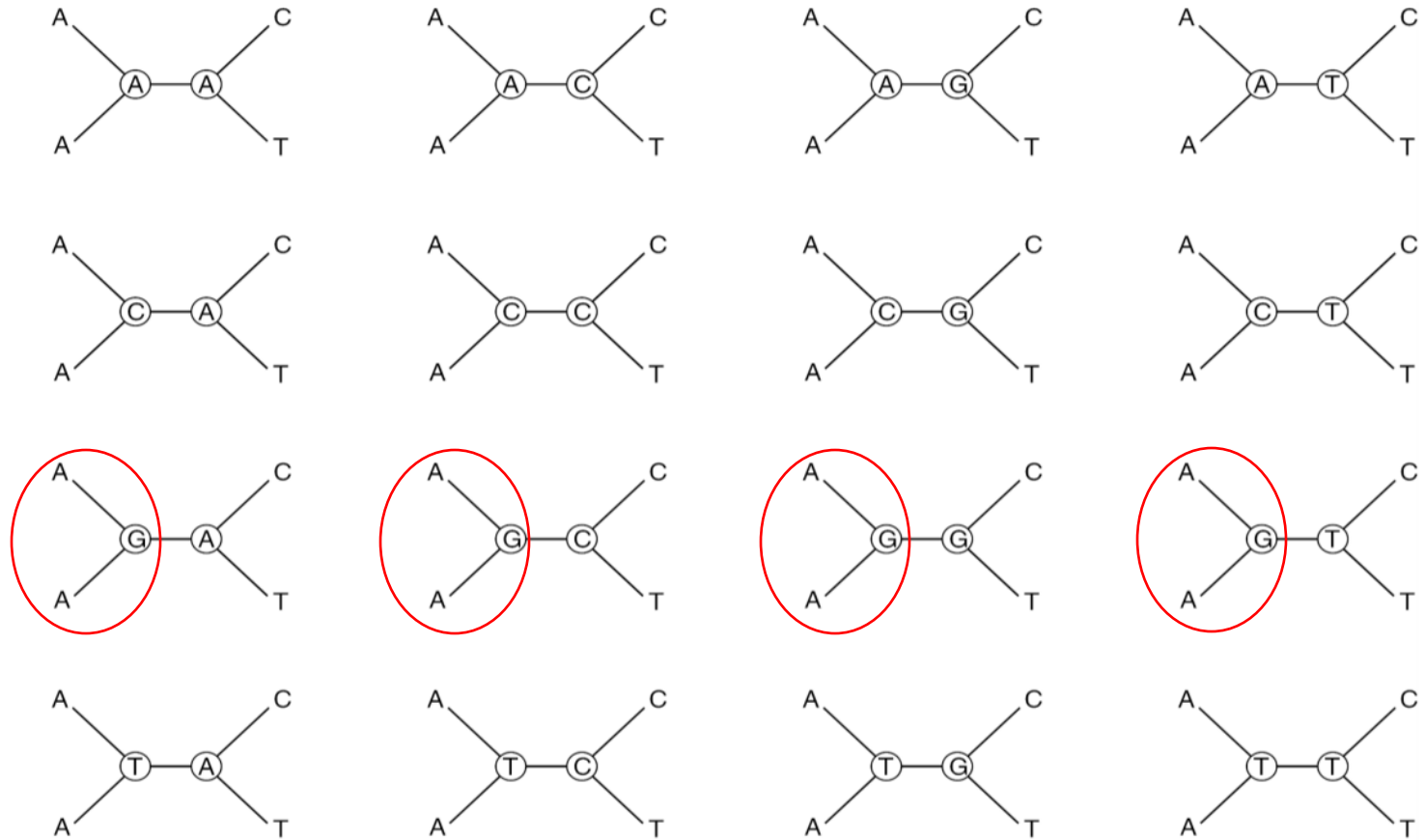
Note use of the OR probability rule

2nd row shared bipartition



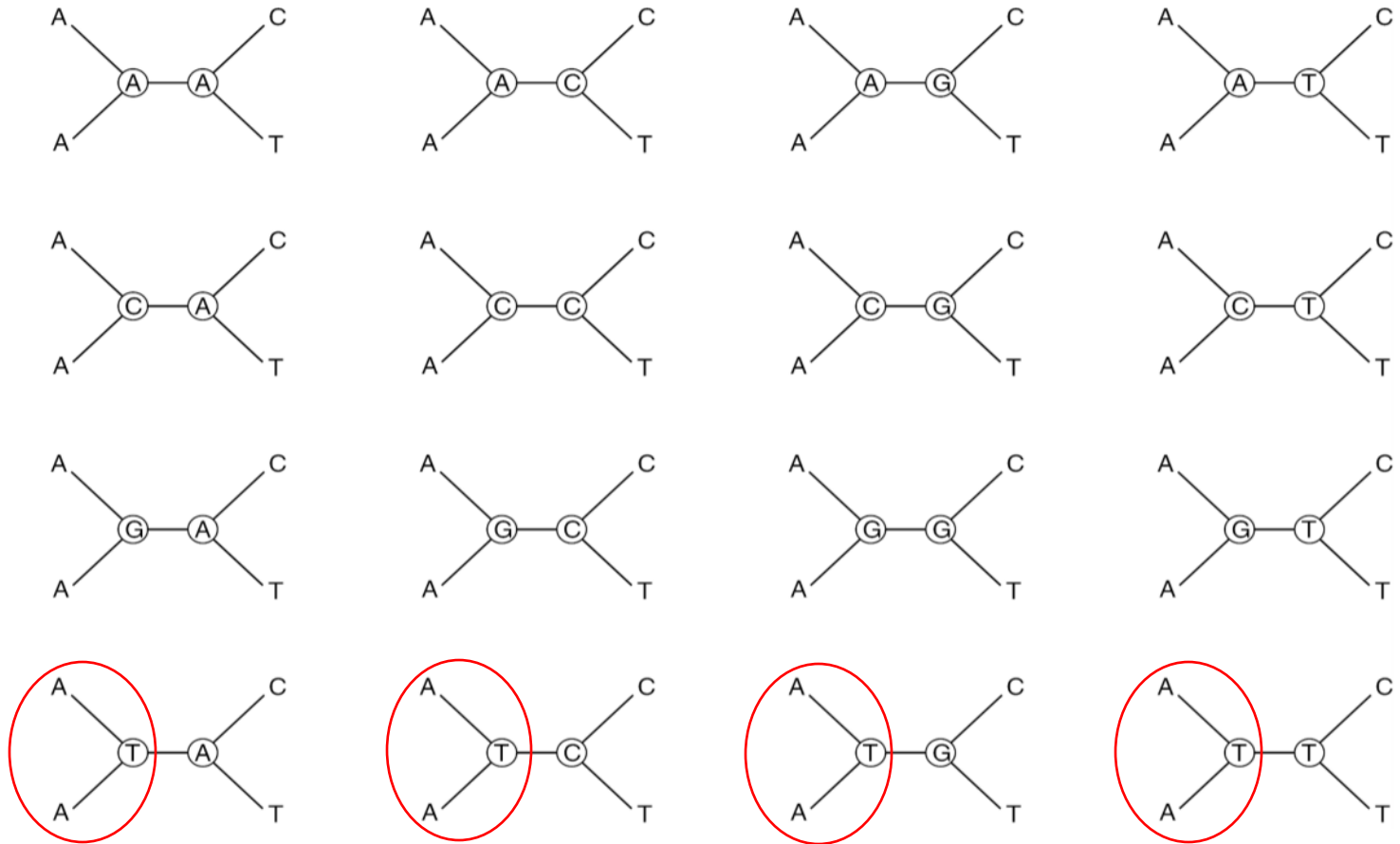
Note use of the OR probability rule

3rd row shared bipartition



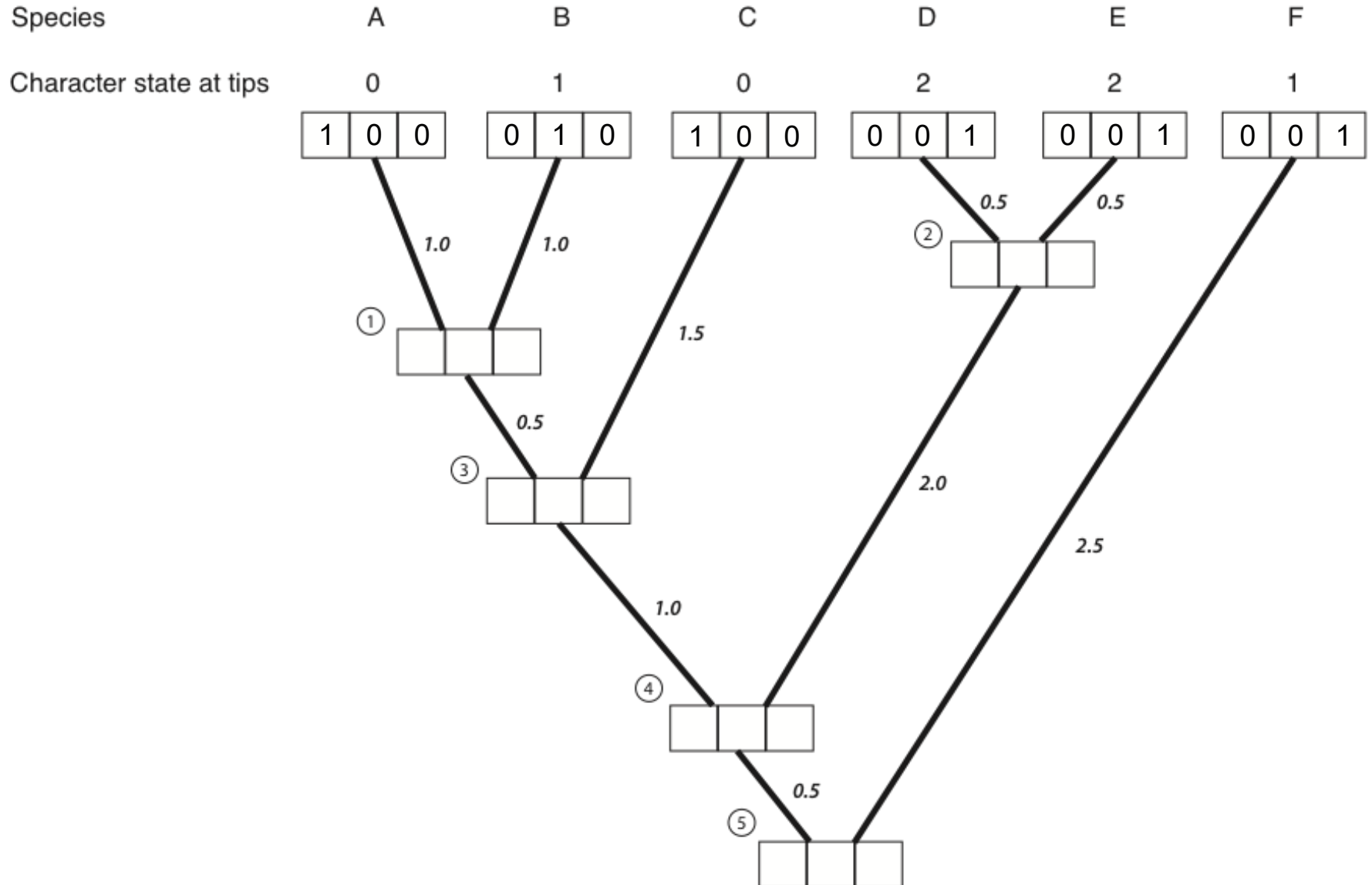
Note use of the OR probability rule

4th row shared bipartition

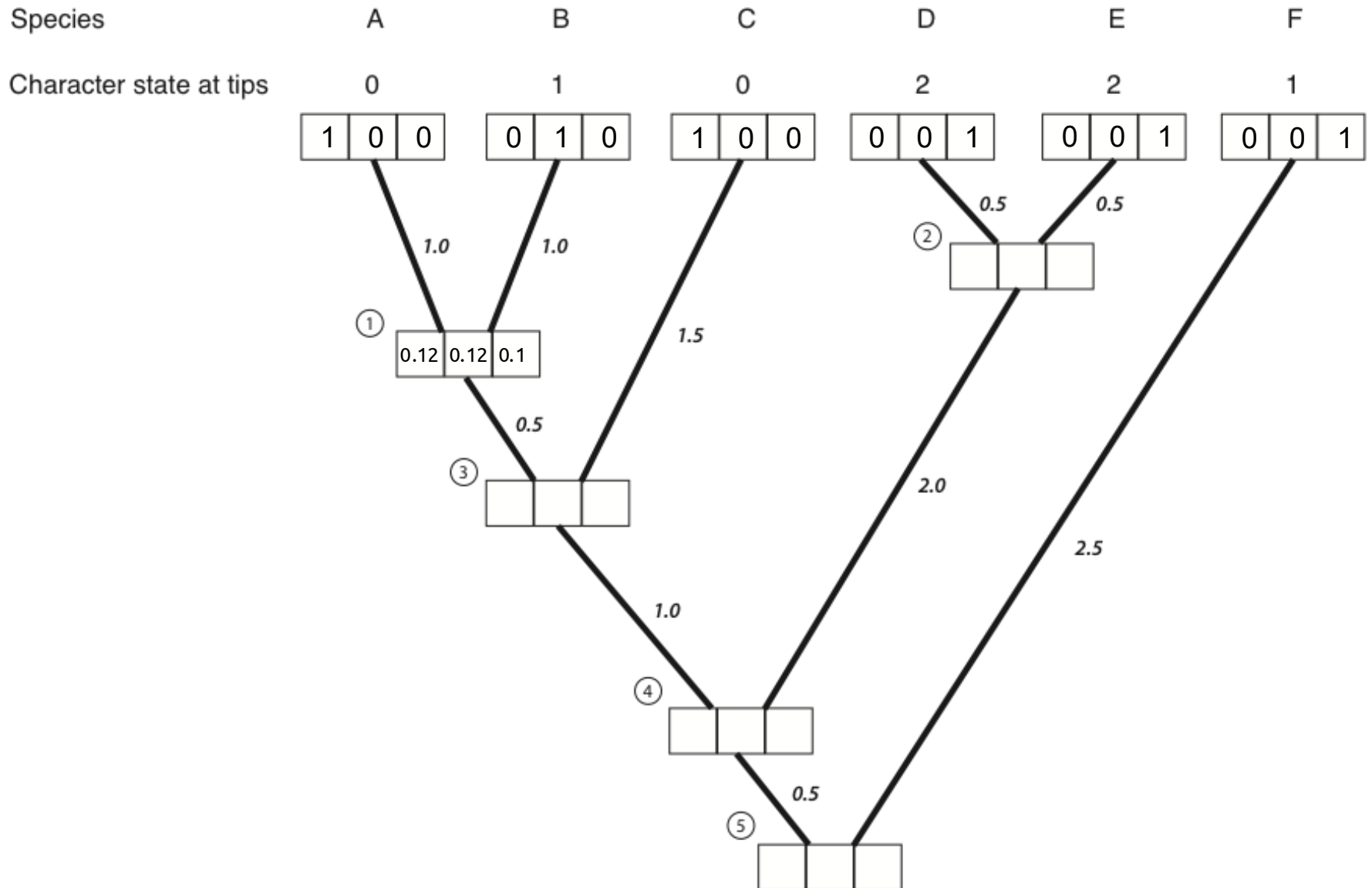


Note use of the OR probability rule

Redundant calculations enable efficient algorithm



Traverse the nodes filling in likelihoods



Traverse the nodes filling in likelihoods

